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(54) **BUILDING DATA PLATFORM WITH
DIGITAL TWIN TRIGGERS AND ACTIONS**

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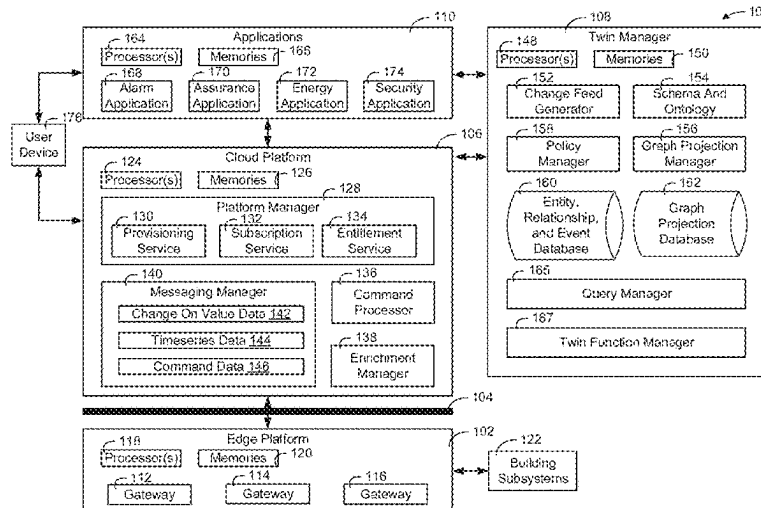
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(57) **ABSTRACT**

A building system including one or more storage devices
storing instructions thereon that, when executed cause the
one or more processors to store a data structure of a digital
twin of an entity of a building, execute a learning algorithm
to construct a trigger rule and an action rule as an input to
the learning algorithm, store the trigger rule and the action
rule in the digital twin, determine that the trigger rule of the
digital twin is triggered by comparing at least some of the
information of the data structure against one or more con-
ditions of the trigger rule by querying the data structure for
data values and comparing the data values against the one or
more conditions of the trigger rule, determine an action of
the action rule of the digital twin responsive to determining
that the trigger rule is triggered by executing the action rule,
and cause one or more devices to operate based on the action
determined by the action rule.

20 Claims, 18 Drawing Sheets



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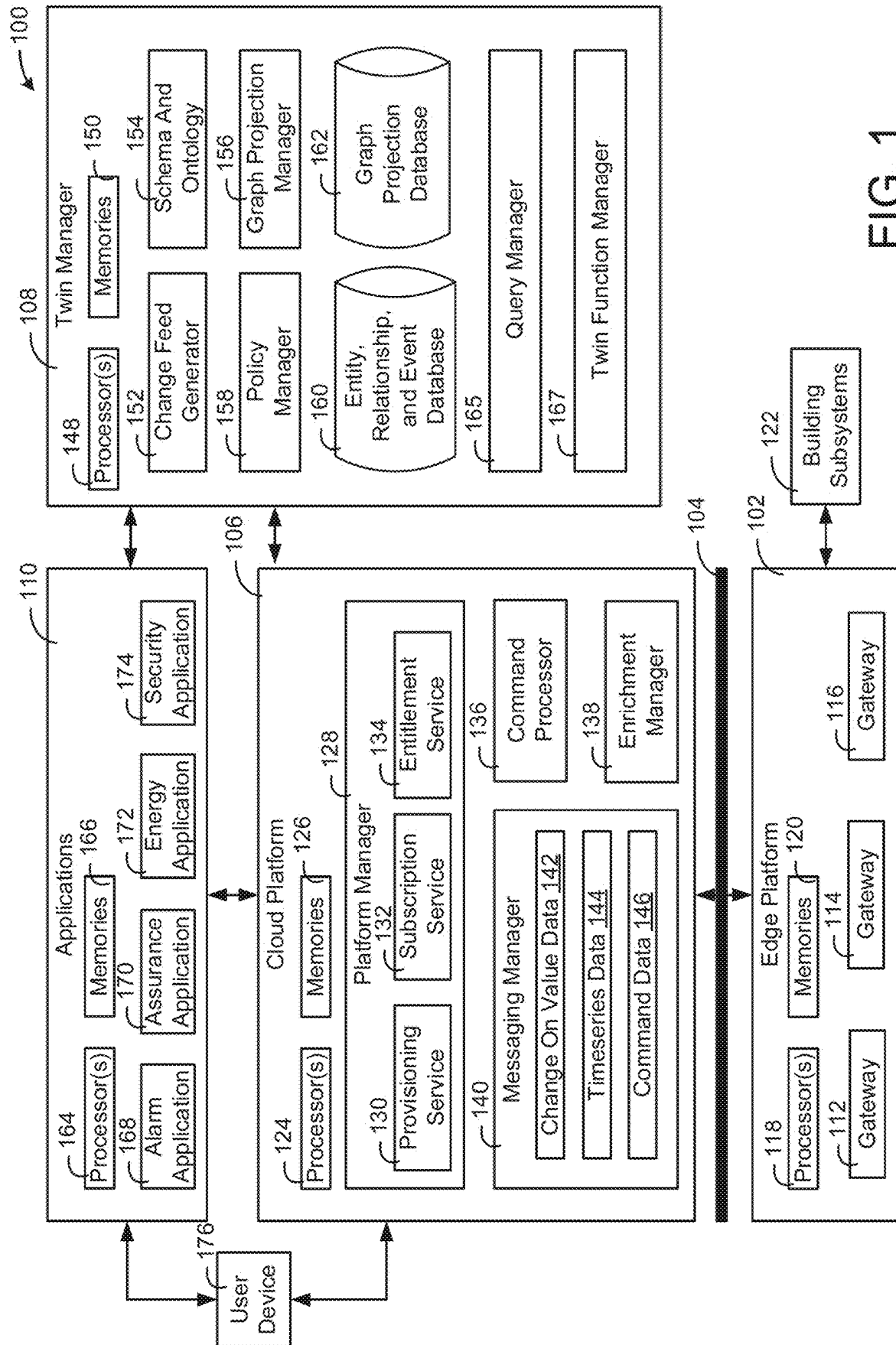
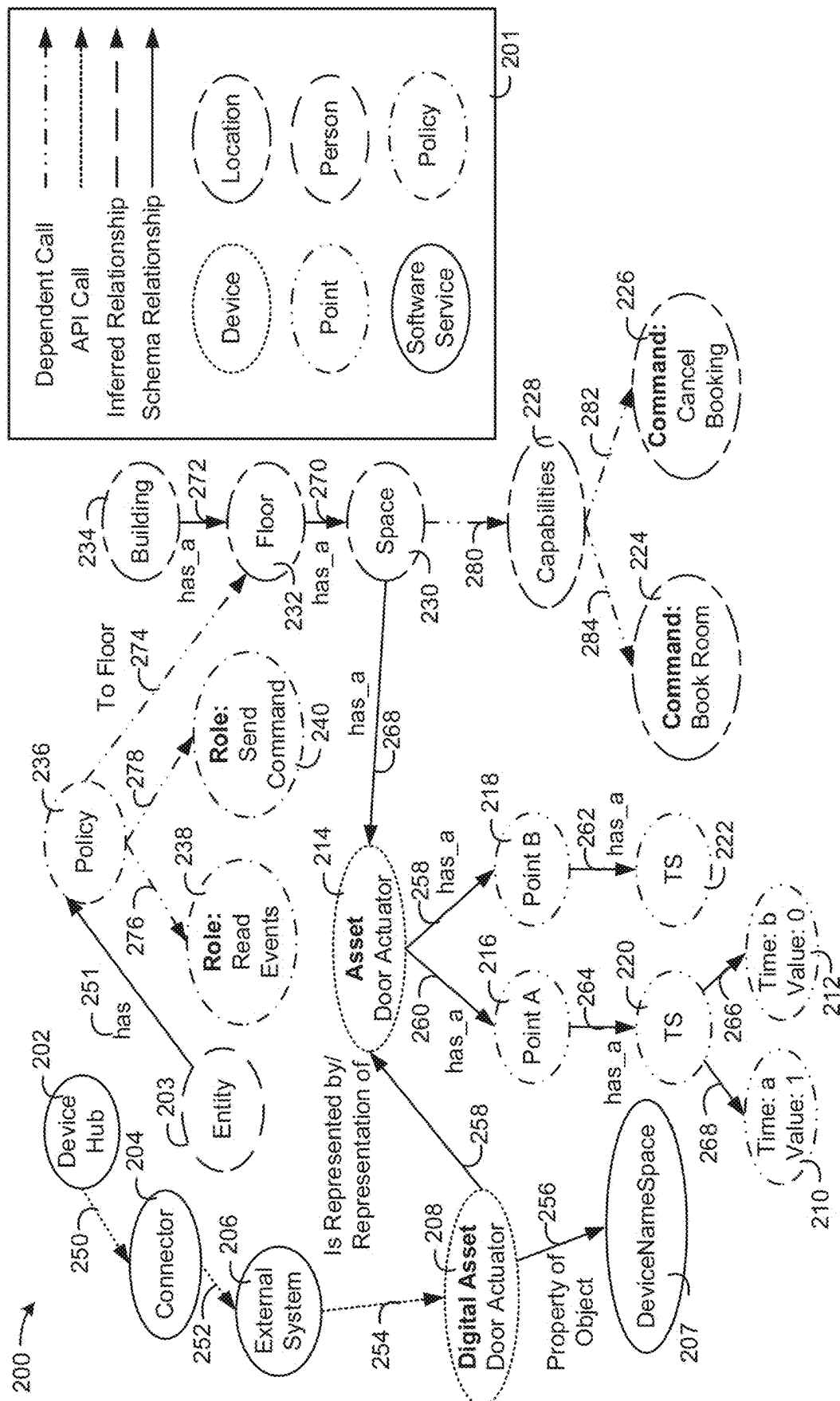
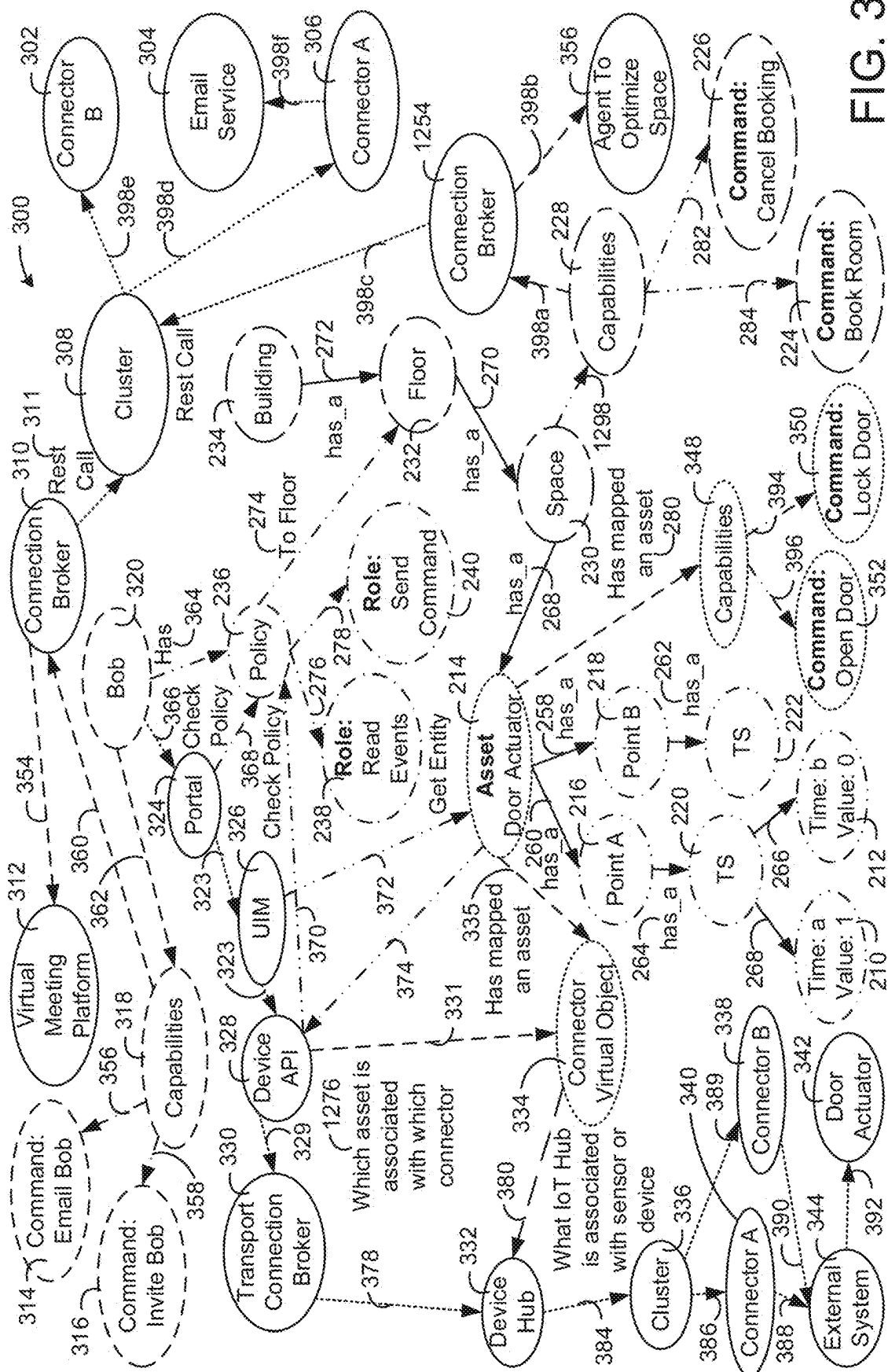


FIG. 1



2016



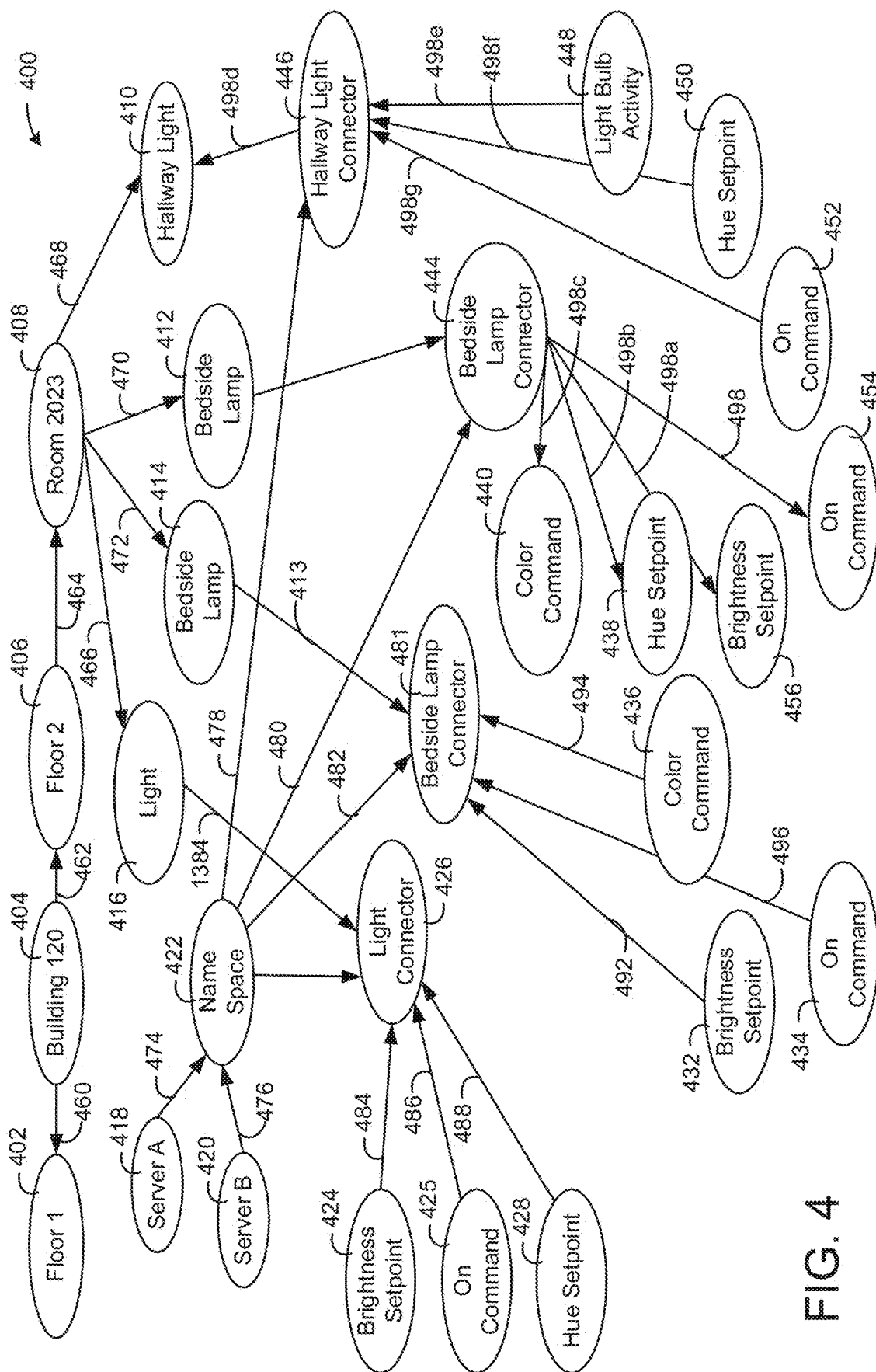
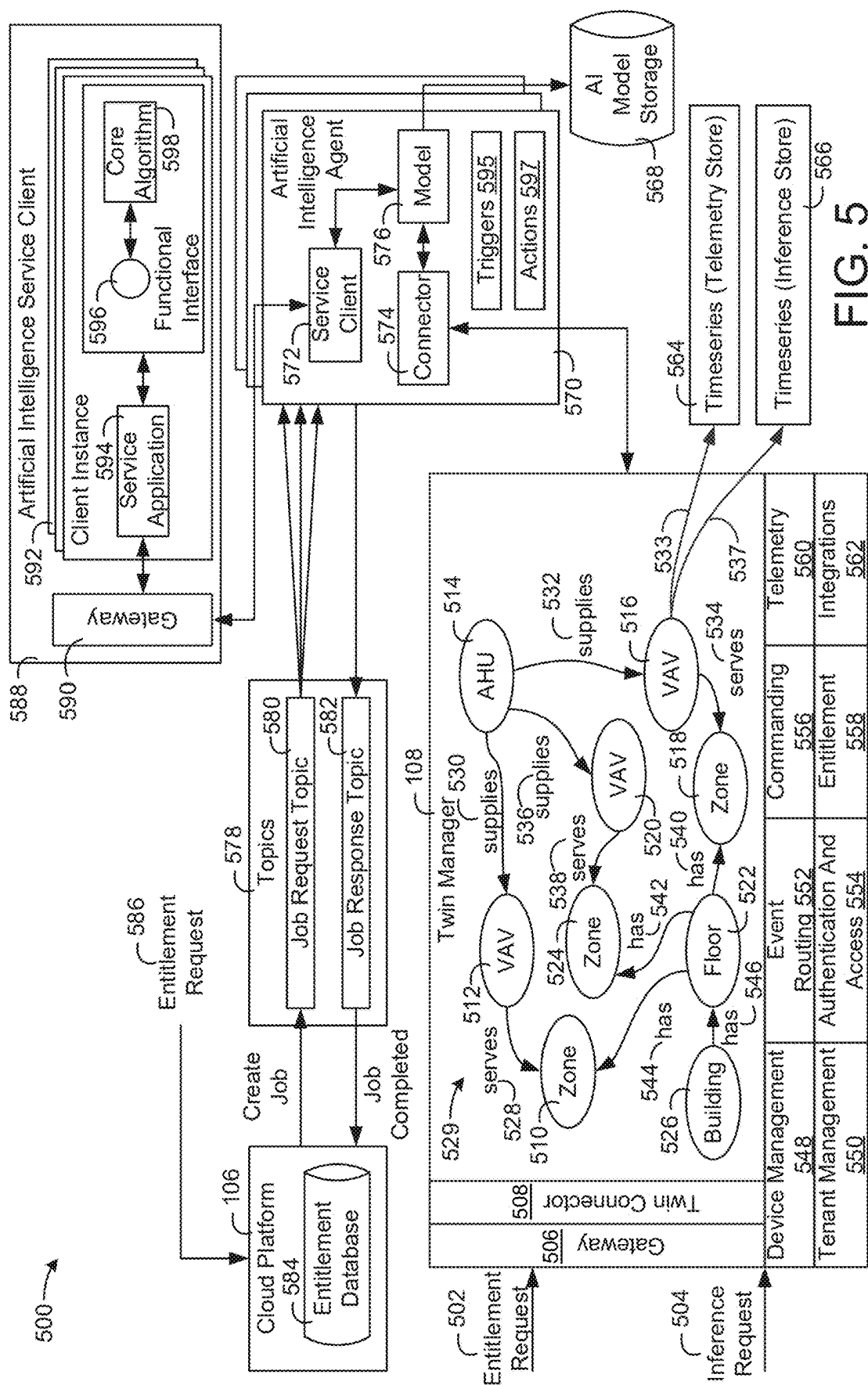


FIG. 4



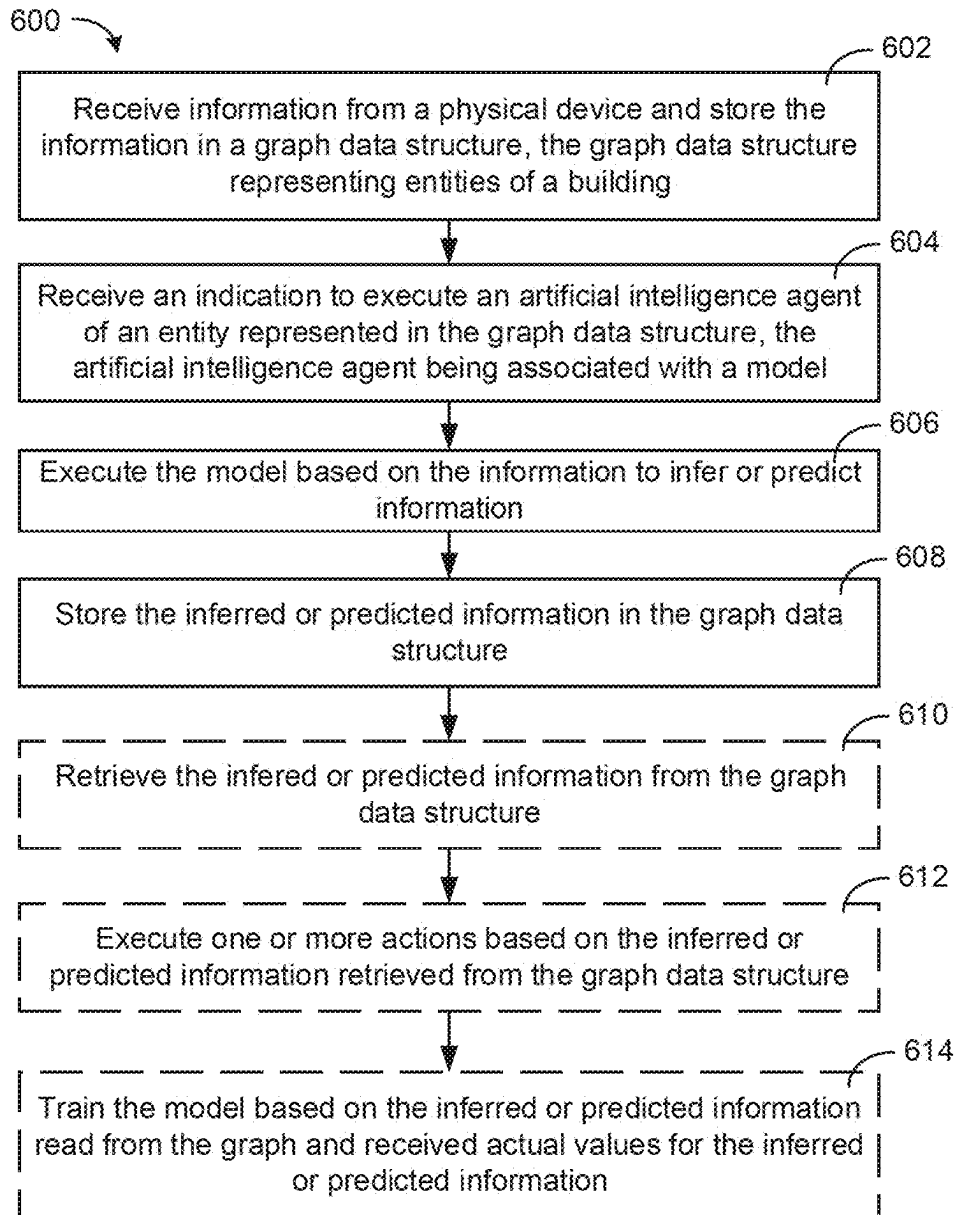


FIG. 6

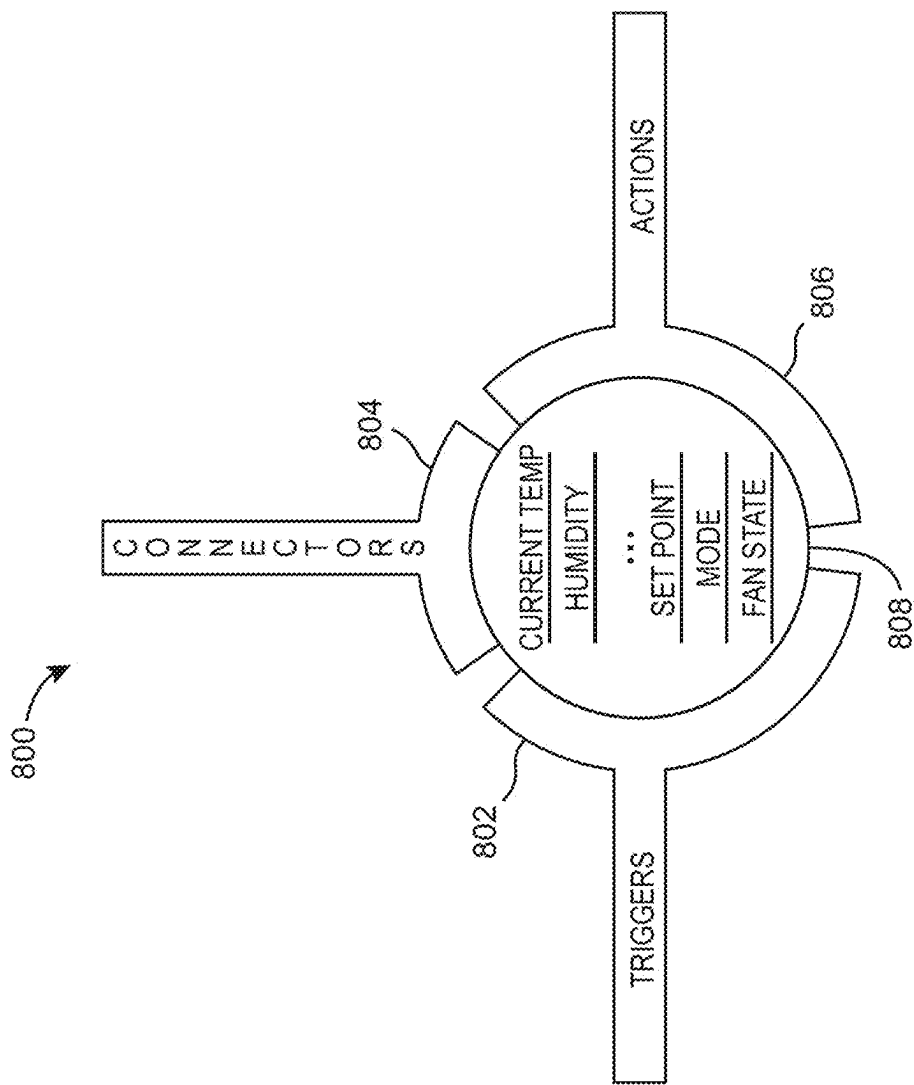


FIG. 7

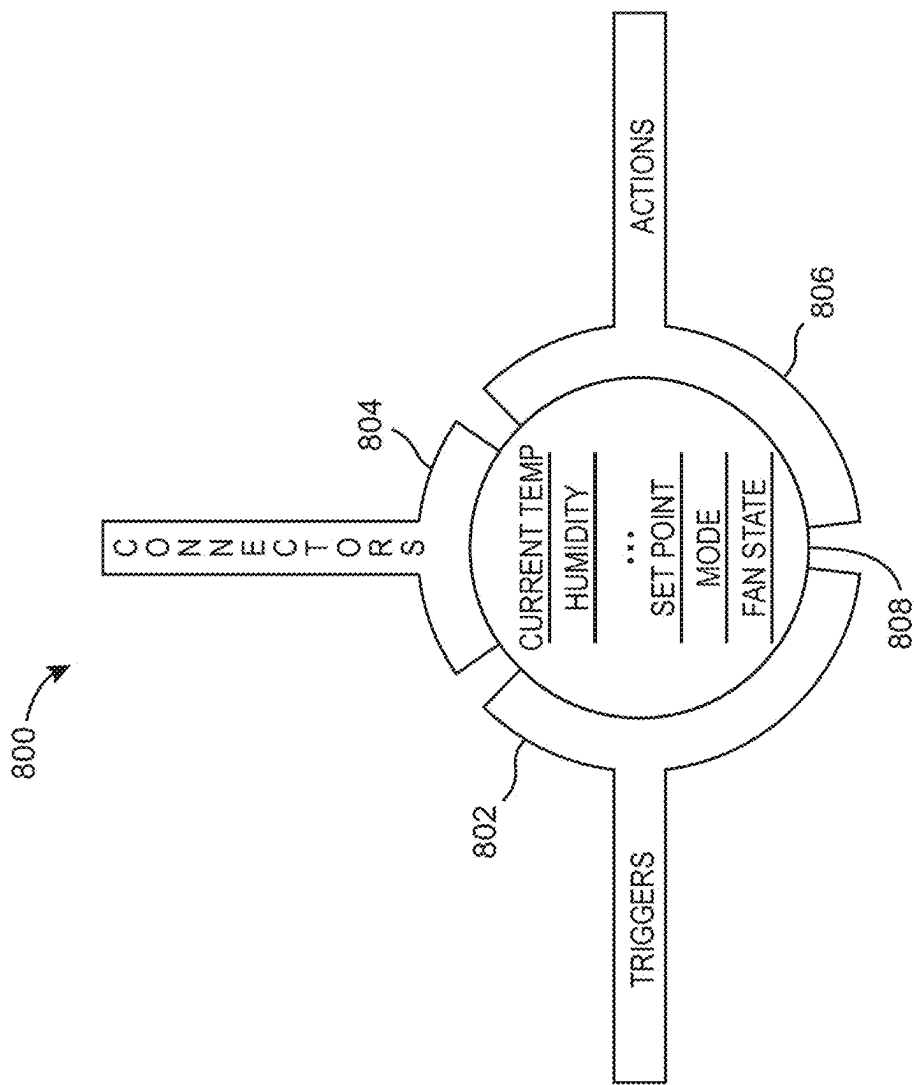


FIG. 8

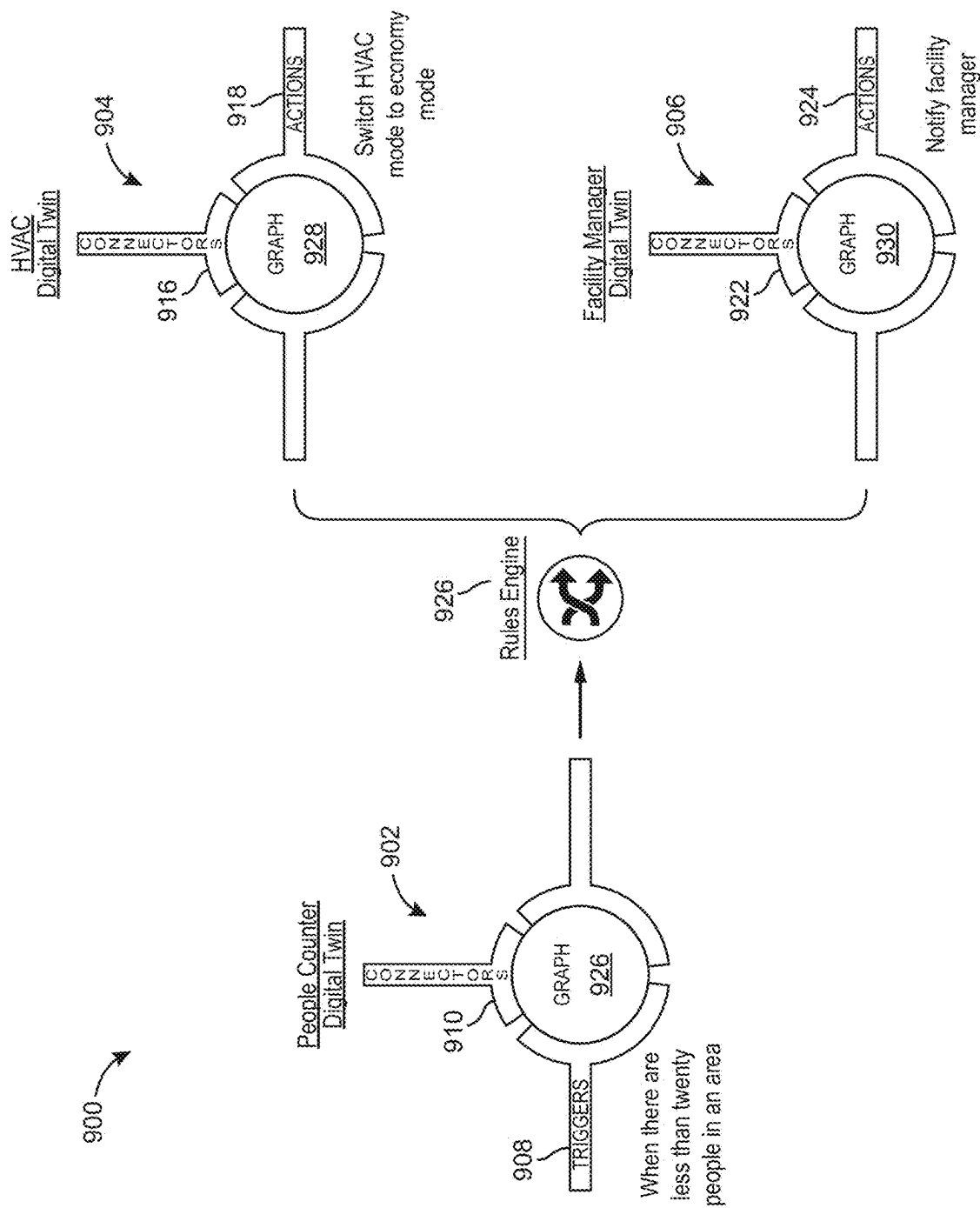


FIG. 9

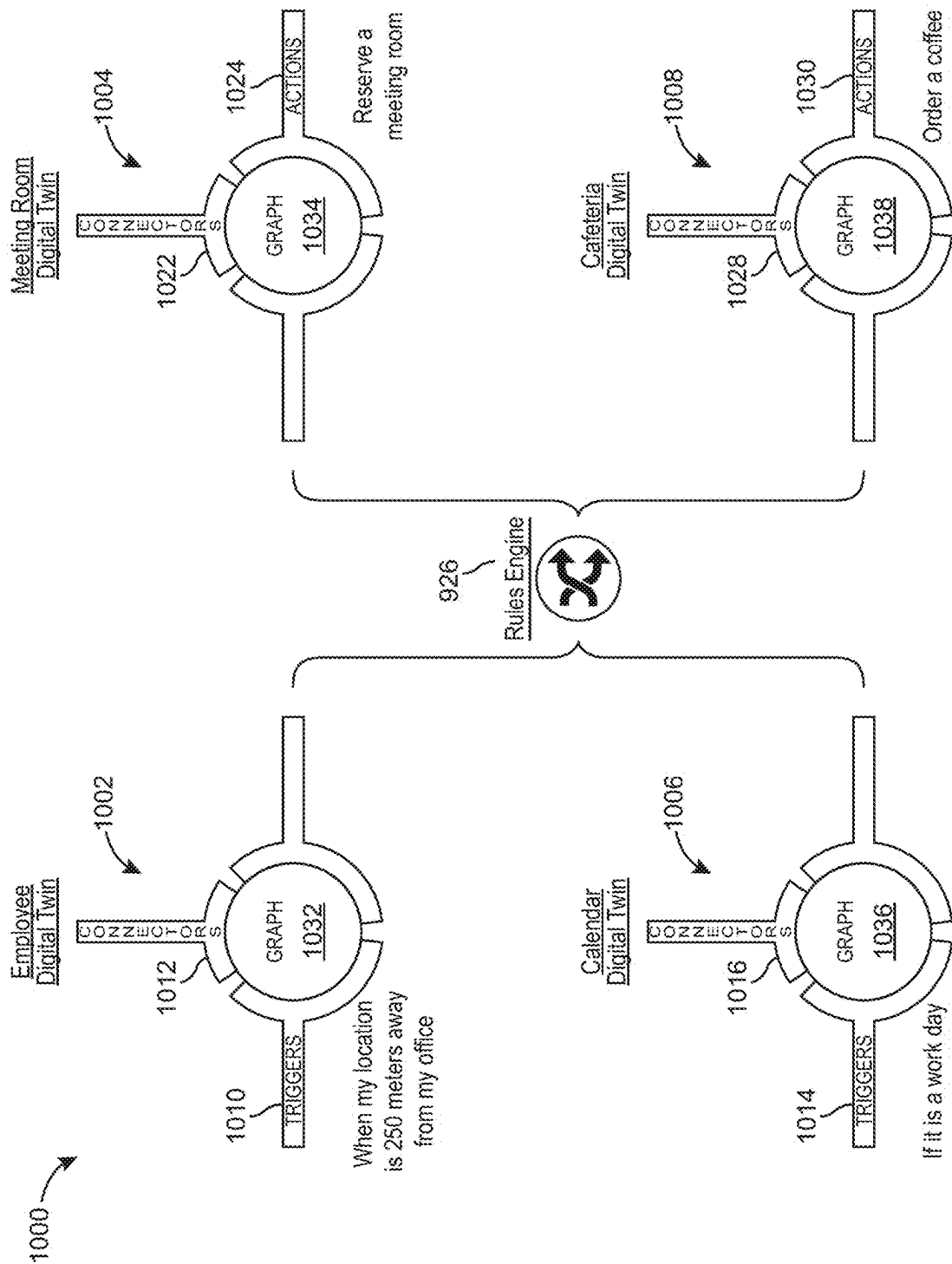


FIG. 10

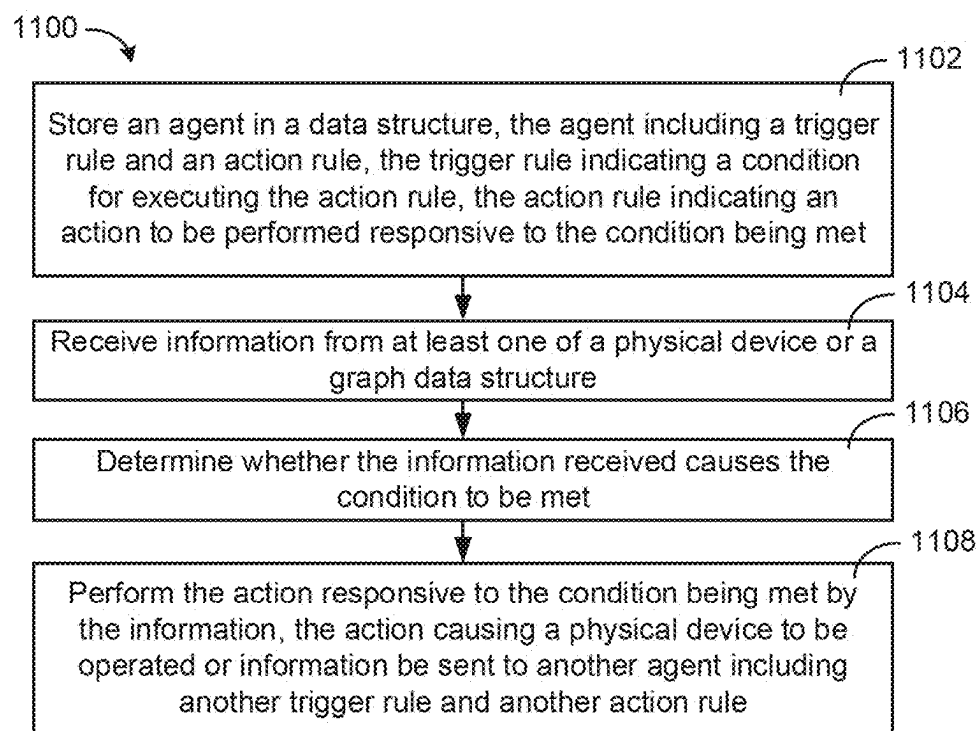
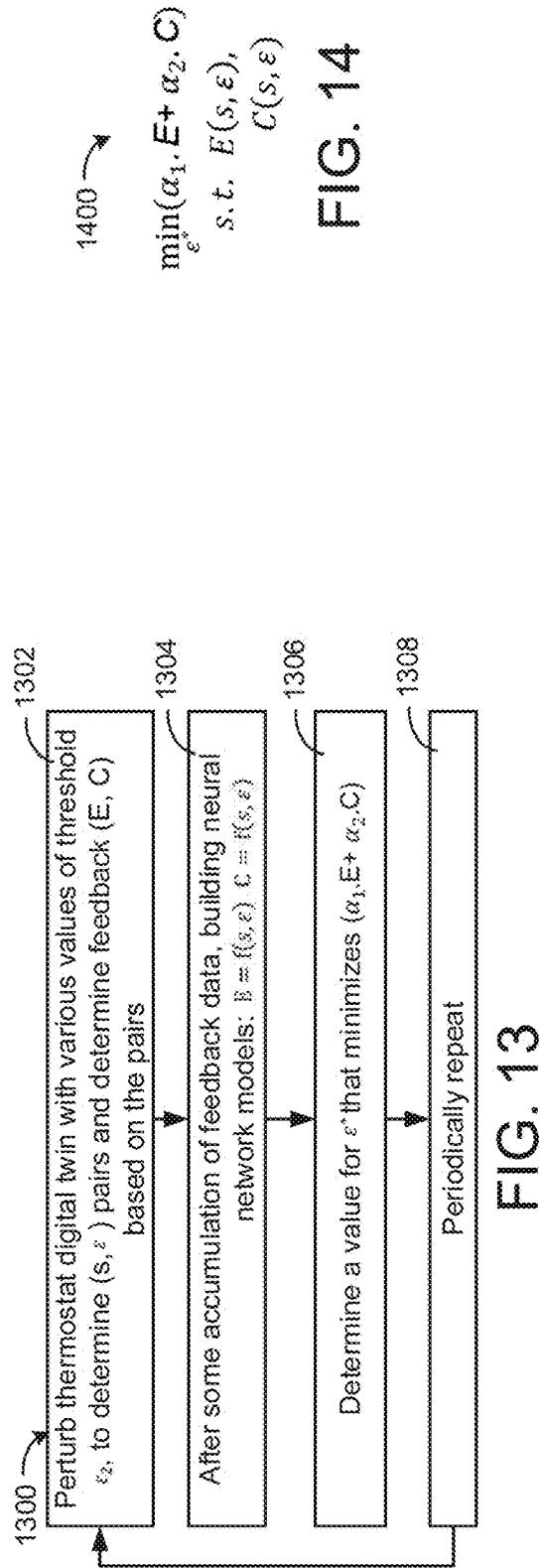
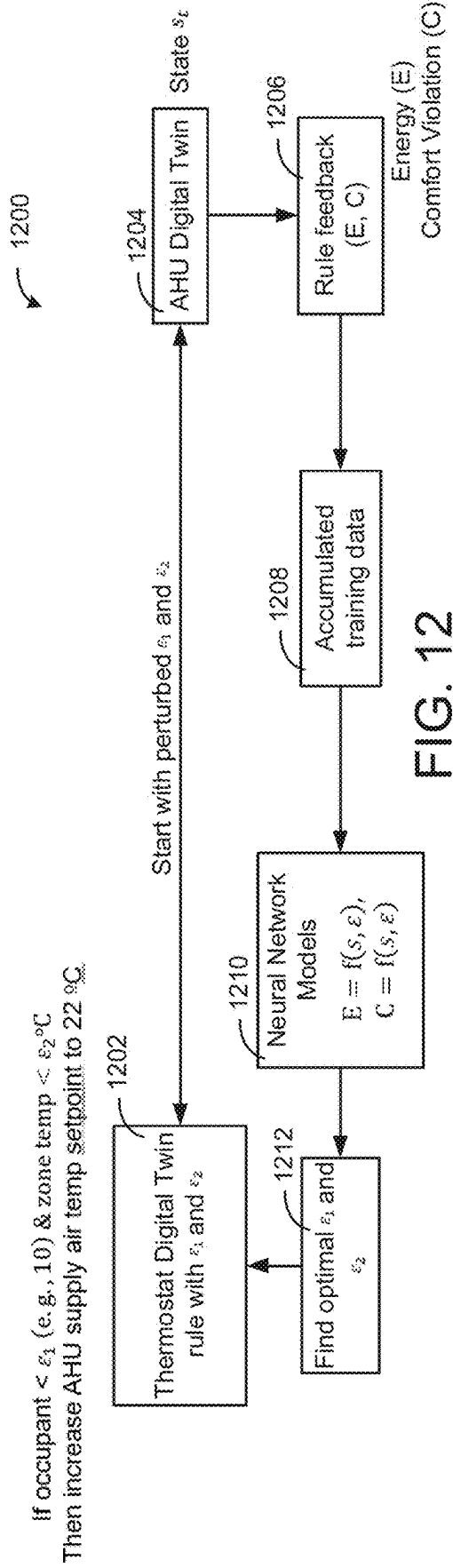


FIG. 11



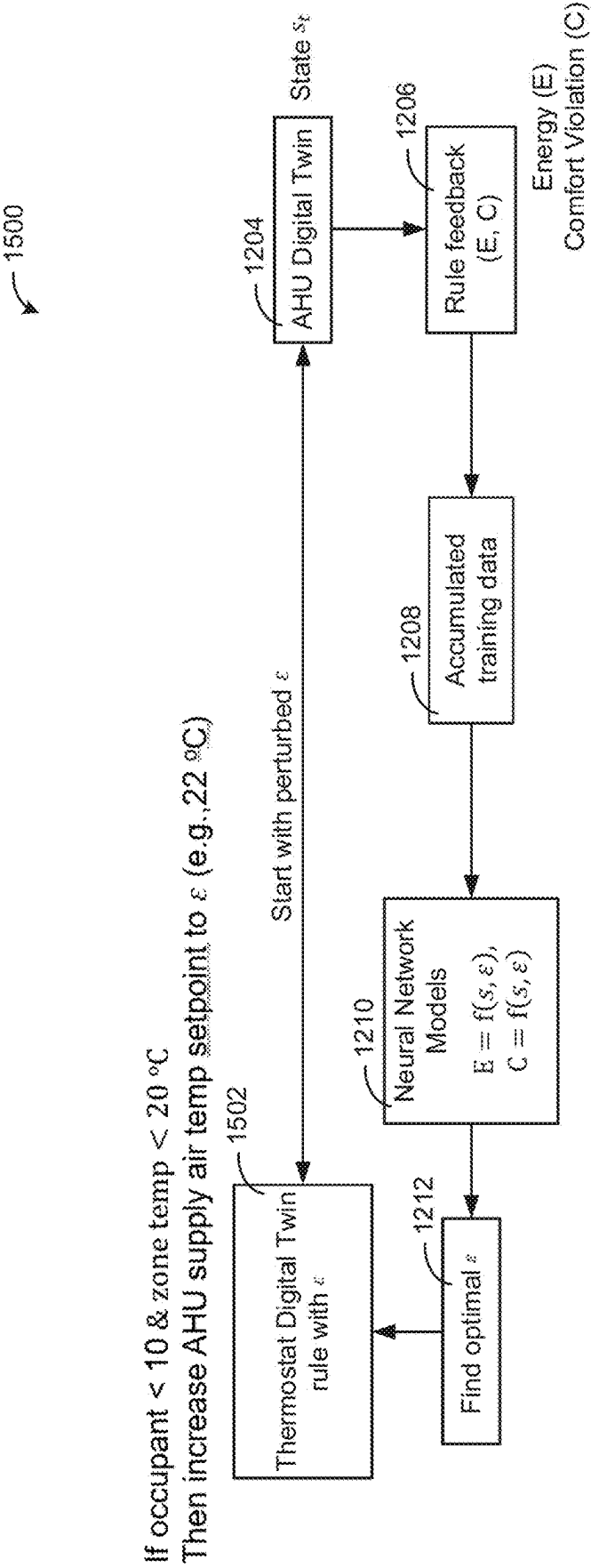


FIG. 15

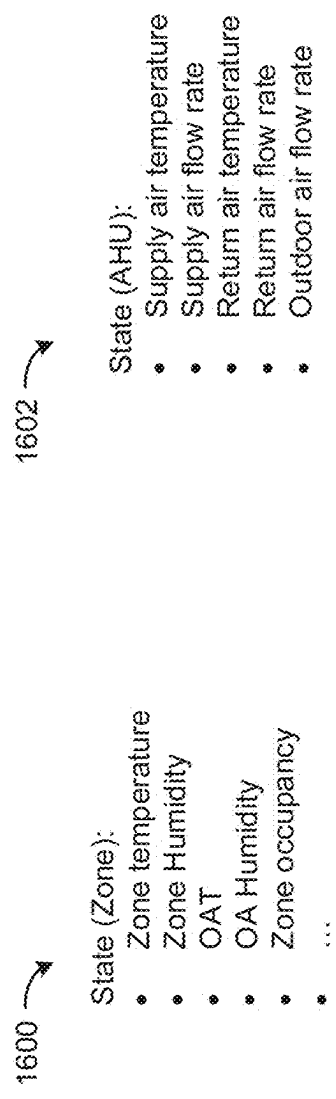
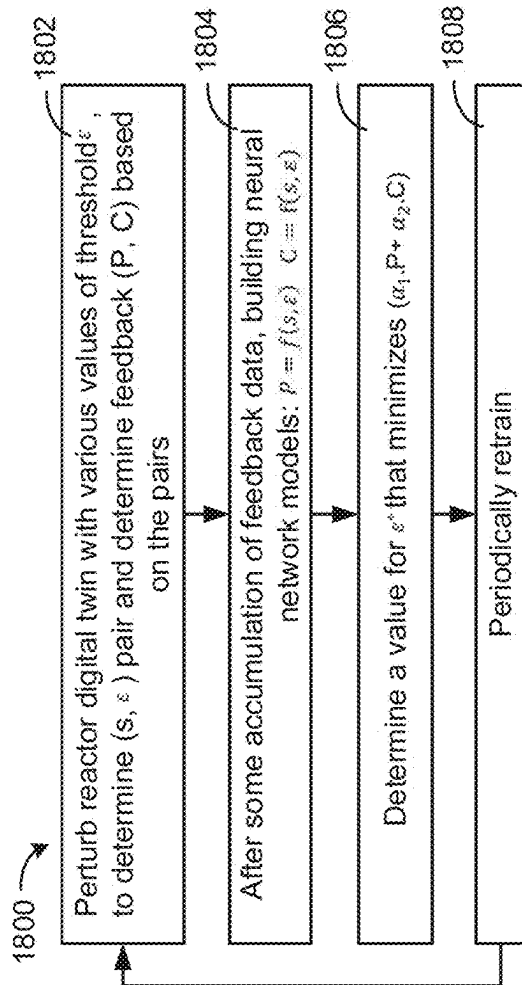
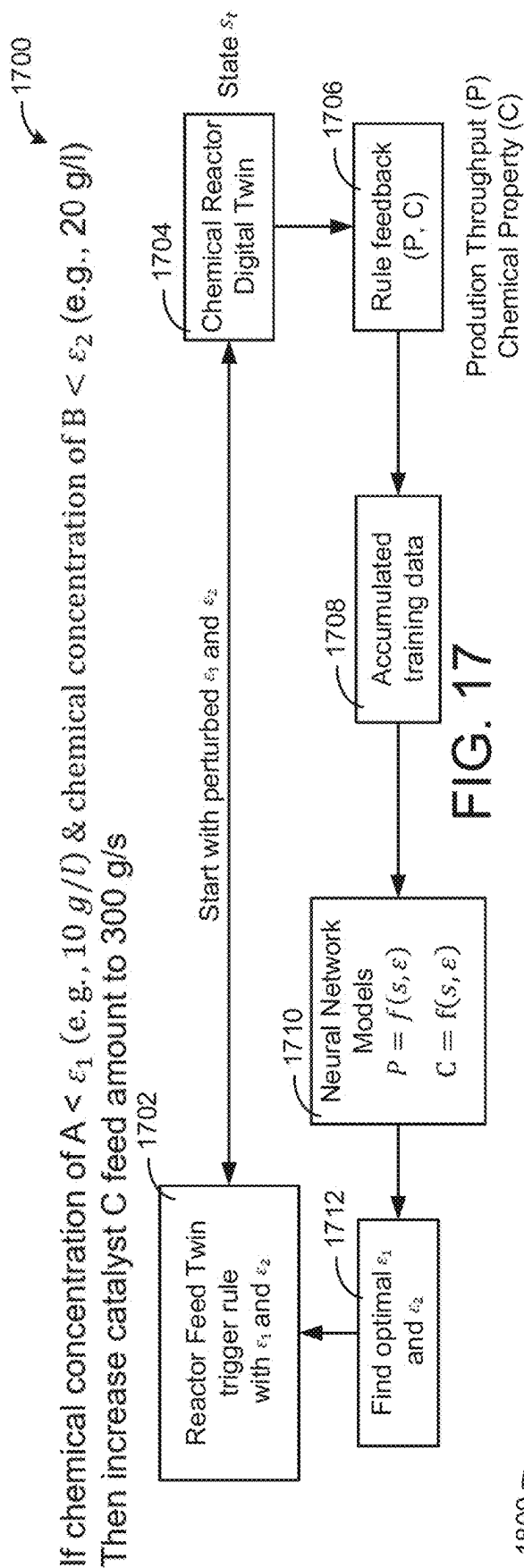


FIG. 16



$$\min_{\varepsilon} (\alpha_1, P + \alpha_2, C)$$

$$s.t. \ E(s, \varepsilon),$$

$$C(s, \varepsilon)$$

FIG. 19

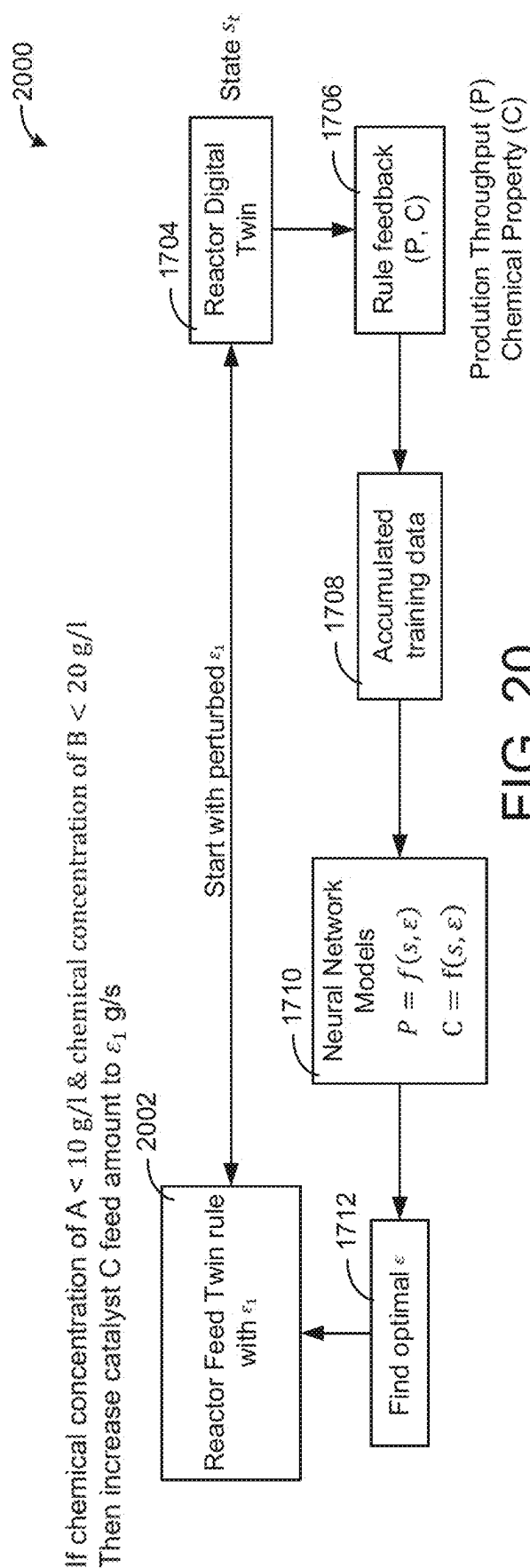


FIG. 20



FIG. 21

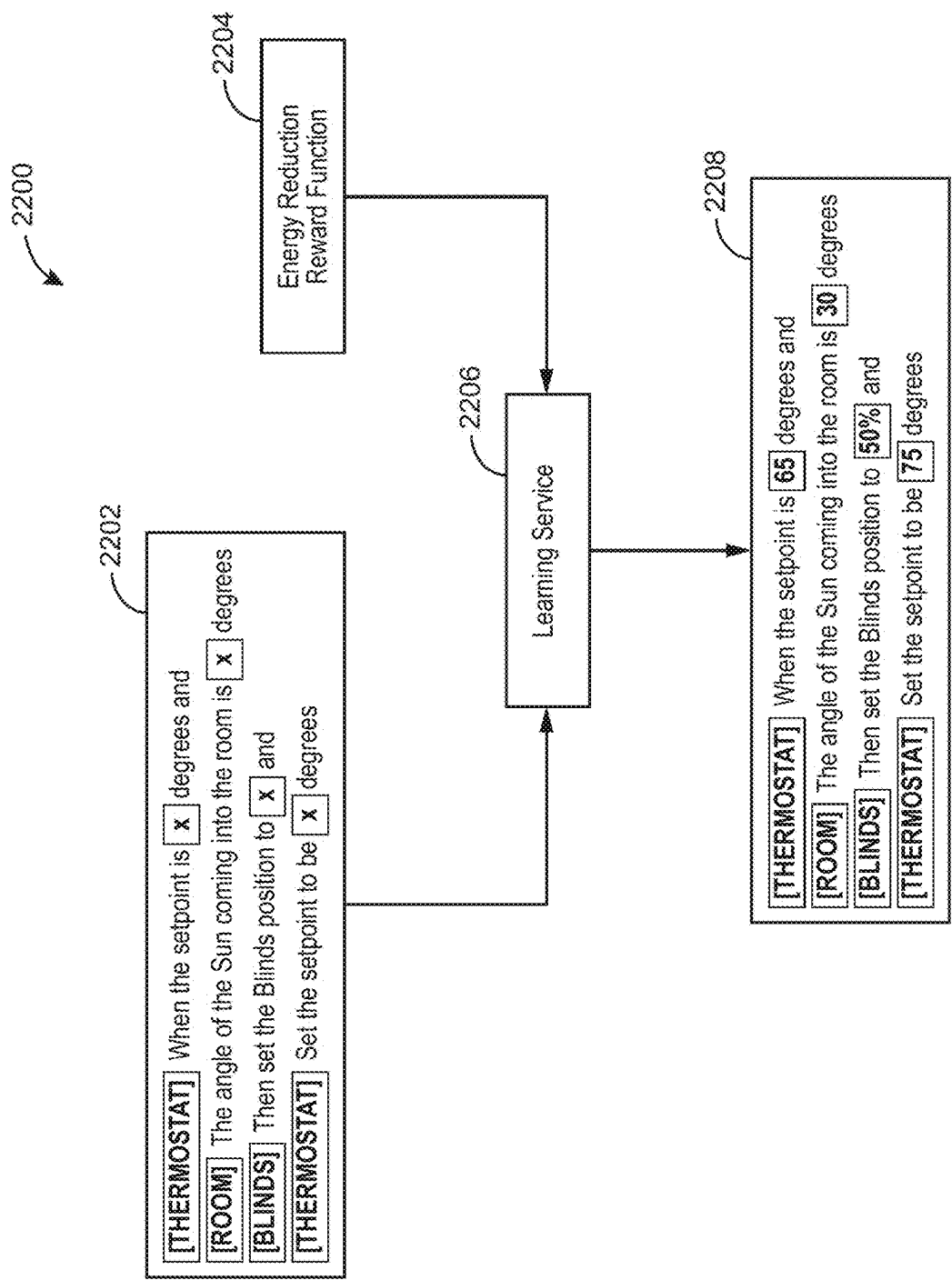


FIG. 22

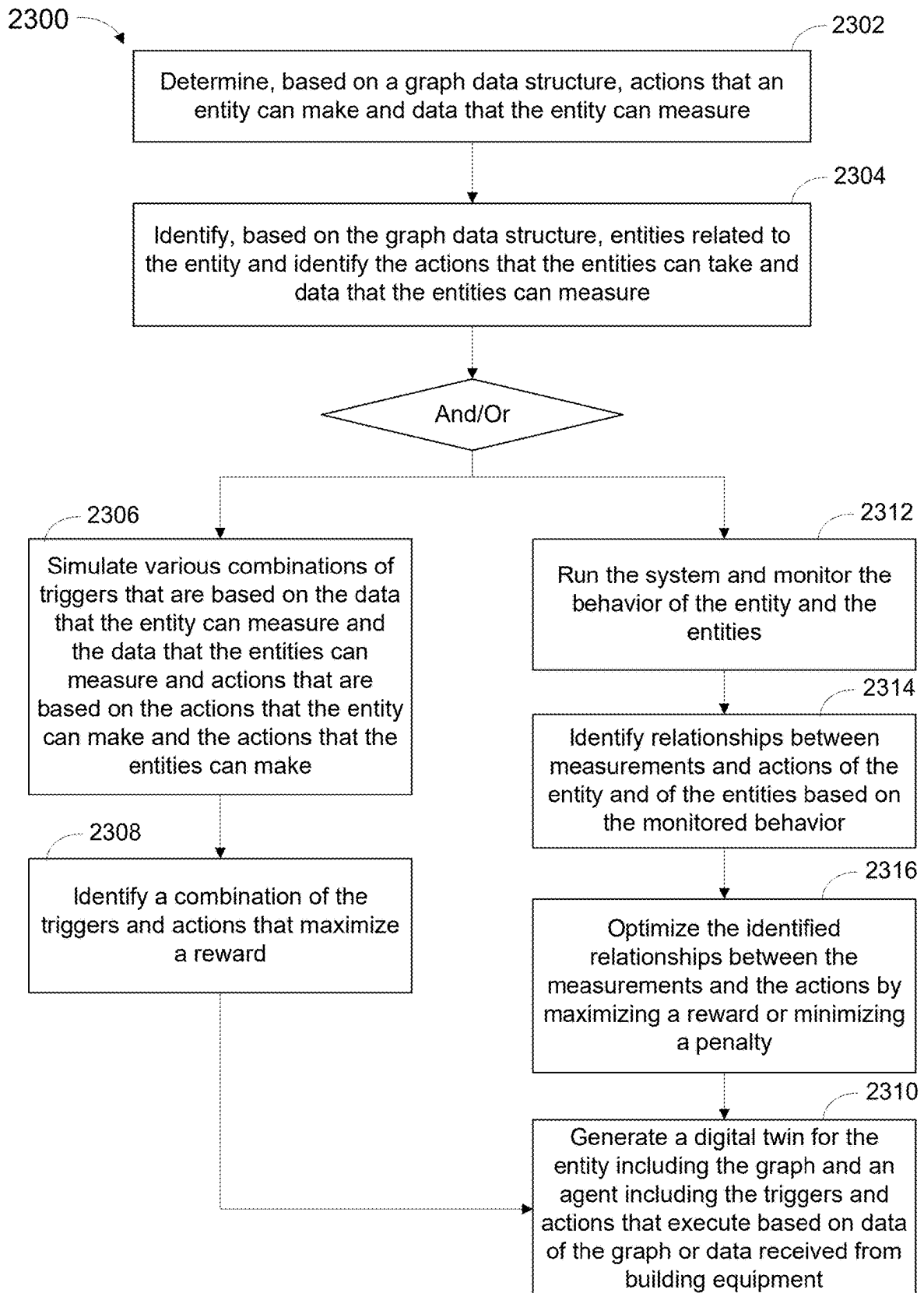


FIG. 23

1

BUILDING DATA PLATFORM WITH DIGITAL TWIN TRIGGERS AND ACTIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/529,120 filed Nov. 17, 2021 which is incorporated herein by reference in its entirety.

BACKGROUND

This application relates generally to a building system of a building. This application relates more particularly to systems for managing and processing data of the building system.

A building may aggregate and store building data received from building equipment and/or other data sources. The building data can be stored in a database. The building can include a building system that operates control algorithms against the data of the database to control the building equipment. However, the development of the control algorithms may be time consuming and require a significant amount of software development. Furthermore, the control algorithms may lack flexibility to adapt to changing circumstances in the building.

SUMMARY

A building system including one or more storage devices storing instructions thereon that, when executed by one or more processors, cause the one or more processors to store a data structure of a digital twin of an entity of a building, the data structure storing information associated with at least one of the entity or a plurality of entities of the building, execute a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm, store the trigger rule and the action rule in the digital twin, determine that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule, determine an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule, and cause one or more devices to operate based on the action determined by the action rule.

In some embodiments, the instructions cause the one or more processors to provide the action to one or more other digital twins, determine that the action triggers one or more trigger rules of the one or more other digital twins, and determine one or more actions of one or more action rules of the one or more other digital twins by executing the one or more action rules responsive to the action triggering the one or more trigger rules.

In some embodiments, the data structure is a graph data structure including nodes and edges between the nodes, wherein the nodes represent the entity, the entities, and data points of the entity and the entities. In some embodiments, the edges represent relationships between the entity, the entities, and the data points of the entity and the entities.

In some embodiments, the instructions cause the one or more processors to cause the one or more devices to operate based on the action by sending a command to the one or more devices to perform the action or store the action in the data structure and retrieve the action from the data structure

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and send the command based on the action retrieved from the data structure to the one or more devices.

In some embodiments, the instructions cause the one or more processors to perform a learning algorithm to learn the trigger rule and the action rule.

In some embodiments, the instructions cause the one or more processors to perform a learning algorithm to learn one or more first parameters for the trigger rule and subsequently learn one or more second parameters for the action rule based at least in part on the one or more first parameters for the trigger rule.

In some embodiments, the instructions cause the one or more processors to learn one or more parameters of the trigger rule or the action rule by adjusting a value of the one or more parameters of the trigger rule or the action rule to different values and simulating one or more states resulting from the different values of the one or more parameters of the trigger rule or the action rule. In some embodiments, the instructions cause the one or more processors to determine one or more updated values for the one or more parameters of the trigger rule or the action rule based on the one or more states resulting from the different values of the one or more parameters of the trigger rule or the action rule.

In some embodiments, the instructions cause the one or more processors to determine the one or more updated value for the one or more parameters of the trigger rule or the action rule by determining a particular reward for the one or more states resulting from the different values of the one or more parameters of the trigger rule or the action rule, training a model that predicts a reward based on the particular reward for the one or more states and the different values of the one or more parameters of the trigger rule or the action rule, and performing an optimization that determines the one or more updated value for the one or more parameters of the trigger rule or the action rule based on the model and the reward.

In some embodiments, the instructions cause the one or more processors to generate the digital twin by identifying actions and data associated with the entity based on the data structure, identifying that the entities are related to the entity based on the data structure, identifying actions and data associated with the entities based on the data structure, and generating the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the entities.

In some embodiments, the instructions cause the one or more processors to generate the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the entities by determining trigger rules for the data of the entity and the data of the entities, determining action rules for the actions of the entity and the actions of the entities, simulating combinations of the trigger rules and action rules and determining rewards for the combinations, and selecting the trigger rule and the action rule from the combinations of the trigger rules and the action rules based on the rewards of the combinations.

In some embodiments, the instructions cause the one or more processors to generate the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the entities by determining trigger rules for the data of the entity and the data of the entities, the trigger rules including the trigger rule, determining action rules for the actions of the entity and the actions of the entities, the action rules including the action rule, monitoring behavior of the building system of the building, determine a relationship between the

trigger rule of the trigger rules and the action rule of the action rules indicated by the behavior of the building system, and selecting the trigger rule from the trigger rules and the action rule from the action rules based on the relationship.

Another implementation of the present disclosure is a method including storing, by a processing circuit, a data structure of a digital twin of an entity of a building, the data structure storing information associated with at least one of the entity or a plurality of entities of the building, executing, by the processing circuit, a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm, storing, by the processing circuit, the trigger rule and the action rule in the digital twin, determining, by the processing circuit, that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule, determining, by the processing circuit, an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule, and causing, by the processing circuit, one or more devices to operate based on the action determined by the action rule.

In some embodiments, the method includes providing, by the processing circuit, the action to one or more other digital twins, determining, by the processing circuit, that the action triggers one or more trigger rules of the one or more other digital twins, and determining, by the processing circuit, one or more actions of one or more action rules of the one or more other digital twins by executing the one or more action rules responsive to the action triggering the one or more trigger rules.

In some embodiments, the data structure is a graph data structure including nodes and edges between the nodes, wherein the nodes represent the entity, the entities, and data points of the entity and the entities. In some embodiments, the edges represent relationships between the entity, the entities, and the data points of the entity and the entities.

In some embodiments, causing, by the processing circuit, the one or more devices to operate based on the action includes sending a command to the one or more devices to perform the action or store the action in the data structure and retrieve the action from the data structure and send the command based on the action retrieved from the data structure to the one or more devices.

In some embodiments, the method includes performing, by the processing circuit, a learning algorithm to learn the trigger rule and the action rule.

In some embodiments, the method includes performing, by the processing circuit, a learning algorithm to learn one or more first parameters for the trigger rule and subsequently learn one or more second parameters for the action rule based at least in part on the one or more first parameters for the trigger rule.

In some embodiments, the method includes learning, by the processing circuit, one or more parameters of the trigger rule or the action rule by adjusting a value of the one or more parameters of the trigger rule or the action rule to different values, simulating one or more states resulting from the different values of the one or more parameters of the trigger rule or the action rule, and determining one or more updated values for the one or more parameters of the trigger rule or the action rule based on the one or more states resulting from the different values of the one or more parameters of the trigger rule or the action rule.

In some embodiments, the method includes generating, by the processing circuit, the digital twin by identifying actions and data associated with the entity based on the data structure, identifying that the entities are related to the entity based on the data structure, identifying actions and data associated with the entities based on the data structure, and generating the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the entities.

Another implementation is one or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations. The operations include storing a data structure of a digital twin of an entity of a building, the data structure storing information associated with at least one of the entity or a plurality of entities of the building, executing a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm, storing the trigger rule and the action rule in the digital twin, determining that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule, determining an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule, and causing one or more devices to operate based on the action determined by the action rule.

In one implementation, the operations further include identifying that a relationship exists between the trigger rule and the action rule and determining the action of the action rule responsive to determining that the trigger rule is triggered and that the relationship exists between the trigger rule and the action rule.

In one implementation, the operations further include storing a second digital twin associated with a second entity of the building, the second digital twin comprising a second action rule, identifying a relationship between the trigger rule of the digital twin and the action rule of the second digital twin, and determining a second action based on the second action rule of the second digital twin responsive to determining that the trigger rule of the digital twin is triggered.

In one implementation, the operations further include: receiving first timeseries data from the entity, receiving second timeseries data from a second entity of the building separate from the entity, and determining that the trigger rule of the digital twin is triggered by comparing the first timeseries data and the second timeseries data against the one or more conditions of the trigger rule.

In one implementation, the one or more non-transitory computer-readable media the operations to generate the digital twin further include: identifying actions and data associated with the entity based on the data structure, identifying that the plurality of entities are related to the entity based on the data structure, identifying actions and data associated with the plurality of entities based on the data structure, and generating the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the plurality of entities.

In one implementation, the one or more non-transitory computer-readable media operations further cause the one or more processors to generate the trigger rule and the action rule based on the actions and the data associated with the

entity and the actions and the data associated with the plurality of entities by: determining trigger rules for the data of the entity and the data of the plurality of entities, determining action rules for the actions of the entity and the actions of the plurality of entities, simulating combinations of the trigger rules and action rules and determining rewards for the combinations, and selecting the trigger rule and the action rule from the combinations of the trigger rules and the action rules based on the rewards of the combinations.

BRIEF DESCRIPTION OF THE DRAWINGS

Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 is a block diagram of a building data platform including an edge platform, a cloud platform, and a twin manager, according to an exemplary embodiment.

FIG. 2 is a graph projection of the twin manager of FIG. 1 including application programming interface (API) data, capability data, policy data, and services, according to an exemplary embodiment.

FIG. 3 is another graph projection of the twin manager of FIG. 1 including application programming interface (API) data, capability data, policy data, and services, according to an exemplary embodiment.

FIG. 4 is a graph projection of the twin manager of FIG. 1 including equipment and capability data for the equipment, according to an exemplary embodiment.

FIG. 5 is a block diagram of a system for managing a digital twin where an artificial intelligence agent can be executed to infer information for an entity of a graph, according to an exemplary embodiment.

FIG. 6 is a flow diagram of a process for executing an artificial intelligence agent to infer and/or predict information, according to an exemplary embodiment.

FIG. 7 is a diagram of a digital twin including a connector and a database, according to an exemplary embodiment.

FIG. 8 is a block diagram of a digital twin including triggers, connectors, actions, and a graph, according to an exemplary embodiment.

FIG. 9 is a block diagram of a people counter digital twin, an HVAC digital twin, and a facility manager digital twin that have triggers and actions that are interconnected, according to an exemplary embodiment.

FIG. 10 is a block diagram of an employee digital twin, a calendar digital twin, a meeting room digital twin, and a cafeteria digital twin that have triggers and actions that are interconnected, according to an exemplary embodiment.

FIG. 11 is a flow diagram an agent of a digital twin executing a trigger rule and an action rule, according to an exemplary embodiment.

FIG. 12 is a block diagram of a trigger rule of a thermostat digital twin where parameters of the trigger rule is trained, according to an exemplary embodiment.

FIG. 13 is a flow diagram of a process for identifying values for the parameters of the trigger rule of FIG. 12, according to an exemplary embodiment.

FIG. 14 is a minimization that can be performed to identify the values for the parameters of the trigger rule of FIGS. 12-13, according to an exemplary embodiment.

FIG. 15 is a block diagram of an action rule of a thermostat digital twin where parameters of the action rule is trained, according to an exemplary embodiment.

FIG. 16 is lists of states of a zone and of an air handler unit that can be used to train the parameters of the trigger rule and the action rule of the thermostat digital twins of FIGS. 12-15, according to an exemplary embodiment.

FIG. 17 is a block diagram of a trigger rule of a chemical reactor digital twin where parameters of the trigger rule are trained, according to an exemplary embodiment.

FIG. 18 is a flow diagram of a process for identifying values for the parameters of the trigger rule of FIG. 17, according to an exemplary embodiment.

FIG. 19 is a minimization that can be performed to identify the values for the parameters of the trigger rule of FIGS. 17-18, according to an exemplary embodiment.

FIG. 20 is a block diagram of an action rule of a chemical reactor digital twin where parameters of the action rule are trained, according to an exemplary embodiment.

FIG. 21 is lists of states of a reactor and a feed of a reactor that can be included in the trigger rule and the action rule of FIGS. 12-15, according to an exemplary embodiment.

FIG. 22 is a block diagram of triggers and actions that can be constructed and learned for a digital twin, according to an exemplary embodiment.

FIG. 23 is a flow diagram of a process for constructing triggers and actions for a digital twin, according to an exemplary embodiment.

DETAILED DESCRIPTION

Overview

Referring generally to the FIGURES, systems and methods for digital twins of a building are shown, according to various exemplary embodiments. A digital twin can be a virtual representing of a building and/or an entity of the building (e.g., space, piece of equipment, occupant, etc.). Furthermore, the digital twin can represent a service performed in a building, e.g., facility management, equipment maintenance, etc.

In some embodiments, the digital twin can include an information data store and a connector. The information data store can store the information describing the entity that the digital twin operates for (e.g., attributes of the entity, measurements associated with the entity, control points or commands of the entity, etc.). In some embodiments, the data store can be a graph including various nodes and edges. The connector can be a software component the provides telemetry from the entity (e.g., physical device) to the information store. Furthermore, the digital twin can include artificial intelligence (AI), e.g., an AI agent. The AI can be one or more machine learning algorithms and/or models that operate based on information of the information data store and outputs information.

In some embodiments, the AI agent for the digital twin can call an AI service to determine inferences about potential future events and/or predict future data values. In some embodiments, the inferences are potential future states. In some embodiments, the inferences predict a timeseries of a data point into the future. The inferences could be inferred indoor temperature for an hour, inferred future air quality from 15 minute air quality readings, etc. In some embodiments, the digital twin can store and inferred information in a graph data store as a node in the graph data store related to an entity that the digital twin represents. In some embodiments, the digital twin, or other digital twins, can operate against the inferred data, e.g., operate to construct and

implement control algorithms for operating equipment of a building based on predicted future data points of the building.

Furthermore, the AI agent can include various triggers and/or actions, conditions that define when and how command and control occurs for an entity. The triggers and actions can be rule based conditional and operational statements that are associated with a specific digital twin, e.g., are stored and executed by an AI agent of the digital twin. In some embodiments, the building system can identify actions and/or triggers (or parameters for the actions and/or triggers) through machine learning algorithms. In some embodiments, the building system can evaluate the conditions/context of the graph and determine and/or modify the triggers and actions of a digital twin.

Referring now to FIG. 1, a building data platform 100 including an edge platform 102, a cloud platform 106, and a twin manager 108 are shown, according to an exemplary embodiment. The edge platform 102, the cloud platform 106, and the twin manager 108 can each be separate services deployed on the same or different computing systems. In some embodiments, the cloud platform 106 and the twin manager 108 are implemented in off premises computing systems, e.g., outside a building. The edge platform 102 can be implemented on-premises, e.g., within the building. However, any combination of on-premises and off-premises components of the building data platform 100 can be implemented.

The building data platform 100 includes applications 110. The applications 110 can be various applications that operate to manage the building subsystems 122. The applications 110 can be remote or on-premises applications (or a hybrid of both) that run on various computing systems. The applications 110 can include an alarm application 168 configured to manage alarms for the building subsystems 122. The applications 110 include an assurance application 170 that implements assurance services for the building subsystems 122. In some embodiments, the applications 110 include an energy application 172 configured to manage the energy usage of the building subsystems 122. The applications 110 include a security application 174 configured to manage security systems of the building.

In some embodiments, the applications 110 and/or the cloud platform 106 interacts with a user device 176. In some embodiments, a component or an entire application of the applications 110 runs on the user device 176. The user device 176 may be a laptop computer, a desktop computer, a smartphone, a tablet, and/or any other device with an input interface (e.g., touch screen, mouse, keyboard, etc.) and an output interface (e.g., a speaker, a display, etc.).

The applications 110, the twin manager 108, the cloud platform 106, and the edge platform 102 can be implemented on one or more computing systems, e.g., on processors and/or memory devices. For example, the edge platform 102 includes processor(s) 118 and memories 120, the cloud platform 106 includes processor(s) 124 and memories 126, the applications 110 include processor(s) 164 and memories 166, and the twin manager 108 includes processor(s) 148 and memories 150.

The processors can be a general purpose or specific purpose processors, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable processing components. The processors may be configured to execute computer code and/or instructions stored in the memories or received from other computer readable media (e.g., CDROM, network storage, a remote server, etc.).

The memories can include one or more devices (e.g., memory units, memory devices, storage devices, etc.) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. The memories can include random access memory (RAM), read-only memory (ROM), hard drive storage, temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. The memories can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. The memories can be communicably connected to the processors and can include computer code for executing (e.g., by the processors) one or more processes described herein.

The edge platform 102 can be configured to provide connection to the building subsystems 122. The edge platform 102 can receive messages from the building subsystems 122 and/or deliver messages to the building subsystems 122. The edge platform 102 includes one or multiple gateways, e.g., the gateways 112-116. The gateways 112-116 can act as a gateway between the cloud platform 106 and the building subsystems 122. The gateways 112-116 can be the gateways described in U.S. Provisional Patent Application No. 62/951,897 filed Dec. 20, 2019, the entirety of which is incorporated by reference herein. In some embodiments, the applications 110 can be deployed on the edge platform 102. In this regard, lower latency in management of the building subsystems 122 can be realized.

The edge platform 102 can be connected to the cloud platform 106 via a network 104. The network 104 can communicatively couple the devices and systems of building data platform 100. In some embodiments, the network 104 is at least one of and/or a combination of a Wi-Fi network, a wired Ethernet network, a ZigBee network, a Bluetooth network, and/or any other wireless network. The network 104 may be a local area network or a wide area network (e.g., the Internet, a building WAN, etc.) and may use a variety of communications protocols (e.g., BACnet, IP, LON, etc.). The network 104 may include routers, modems, servers, cell towers, satellites, and/or network switches. The network 104 may be a combination of wired and wireless networks.

The cloud platform 106 can be configured to facilitate communication and routing of messages between the applications 110, the twin manager 108, the edge platform 102, and/or any other system. The cloud platform 106 can include a platform manager 128, a messaging manager 140, a command processor 136, and an enrichment manager 138. In some embodiments, the cloud platform 106 can facilitate messaging between the building data platform 100 via the network 104.

The messaging manager 140 can be configured to operate as a transport service that controls communication with the building subsystems 122 and/or any other system, e.g., managing commands to devices (C2D), commands to connectors (C2C) for external systems, commands from the device to the cloud (D2C), and/or notifications. The messaging manager 140 can receive different types of data from the applications 110, the twin manager 108, and/or the edge platform 102. The messaging manager 140 can receive change on value data 142, e.g., data that indicates that a value of a point has changed. The messaging manager 140 can receive timeseries data 144, e.g., a time correlated series of data entries each associated with a particular time stamp.

Furthermore, the messaging manager **140** can receive command data **146**. All of the messages handled by the cloud platform **106** can be handled as an event, e.g., the data **142-146** can each be packaged as an event with a data value occurring at a particular time (e.g., a temperature measurement made at a particular time).

The cloud platform **106** includes a command processor **136**. The command processor **136** can be configured to receive commands to perform an action from the applications **110**, the building subsystems **122**, the user device **176**, etc. The command processor **136** can manage the commands, determine whether the commanding system is authorized to perform the particular commands, and communicate the commands to the commanded system, e.g., the building subsystems **122** and/or the applications **110**. The commands could be a command to change an operational setting that control environmental conditions of a building, a command to run analytics, etc.

The cloud platform **106** includes an enrichment manager **138**. The enrichment manager **138** can be configured to enrich the events received by the messaging manager **140**. The enrichment manager **138** can be configured to add contextual information to the events. The enrichment manager **138** can communicate with the twin manager **108** to retrieve the contextual information. In some embodiments, the contextual information is an indication of information related to the event. For example, if the event is a timeseries temperature measurement of a thermostat, contextual information such as the location of the thermostat (e.g., what room), the equipment controlled by the thermostat (e.g., what VAV), etc. can be added to the event. In this regard, when a consuming application, e.g., one of the applications **110** receives the event, the consuming application can operate based on the data of the event, the temperature measurement, and also the contextual information of the event.

The enrichment manager **138** can solve a problem that when a device produces a significant amount of information, the information may contain simple data without context. An example might include the data generated when a user scans a badge at a badge scanner of the building subsystems **122**. This physical event can generate an output event including such information as "DeviceBadgeScannerID," "BadgeID," and/or "Date/Time." However, if a system sends this data to a consuming application, e.g., Consumer A and a Consumer B, each customer may need to call the building data platform knowledge service to query information with queries such as, "What space, build, floor is that badge scanner in?" or "What user is associated with that badge?"

By performing enrichment on the data feed, a system can be able to perform inferences on the data. A result of the enrichment may be transformation of the message "DeviceBadgeScannerId, BadgeId, Date/Time," to "Region, Building, Floor, Asset, DeviceId, BadgeId, UserName, EmployeeId, Date/Time Scanned." This can be a significant optimization, as a system can reduce the number of calls by $1/n$, where n is the number of consumers of this data feed.

By using this enrichment, a system can also have the ability to filter out undesired events. If there are 100 building in a campus that receive 100,000 events per building each hour, but only 1 building is actually commissioned, only $1/100$ of the events are enriched. By looking at what events are enriched and what events are not enriched, a system can do traffic shaping of forwarding of these events to reduce the cost of forwarding events that no consuming application wants or reads.

An example of an event received by the enrichment manager **138** may be:

```
{
  "id": "someguid",
  "eventType": "Device_Heartbeat",
  "eventTime": "2018-01-27T00:00:00+00:00"
  "eventValue": 1,
  "deviceId": "someguid"
}
```

An example of an enriched event generated by the enrichment manager **138** may be:

```
{
  "id": "someguid",
  "eventType": "Device_Heartbeat",
  "eventTime": "2018-01-27T00:00:00+00:00"
  "eventValue": 1,
  "deviceId": "someguid",
  "buildingName": "Building-48",
  "buildingID": "SomeGuid",
  "panelID": "SomeGuid",
  "panelName": "Building-48-Panel-13",
  "cityID": 371,
  "cityName": "Milwaukee",
  "stateID": 48,
  "stateName": "Wisconsin (WI)",
  "countryID": 1,
  "countryName": "United States"
}
```

By receiving enriched events, an application of the applications **110** can be able to populate and/or filter what events are associated with what areas. Furthermore, user interface generating applications can generate user interfaces that include the contextual information based on the enriched events.

The cloud platform **106** includes a platform manager **128**. The platform manager **128** can be configured to manage the users and/or subscriptions of the cloud platform **106**. For example, what subscribing building, user, and/or tenant utilizes the cloud platform **106**. The platform manager **128** includes a provisioning service **130** configured to provision the cloud platform **106**, the edge platform **102**, and the twin manager **108**. The platform manager **128** includes a subscription service **132** configured to manage a subscription of the building, user, and/or tenant while the entitlement service **134** can track entitlements of the buildings, users, and/or tenants.

The twin manager **108** can be configured to manage and maintain a digital twin. The digital twin can be a digital representation of the physical environment, e.g., a building. The twin manager **108** can include a change feed generator **152**, a schema and ontology **154**, a projection manager **156**, a policy manager **158**, an entity, relationship, and event database **160**, and a graph projection database **162**.

The graph projection manager **156** can be configured to construct graph projections and store the graph projections in the graph projection database **162**. Examples of graph projections are shown in FIGS. **11-13**. Entities, relationships, and events can be stored in the database **160**.

The graph projection manager **156** can retrieve entities, relationships, and/or events from the database **160** and construct a graph projection based on the retrieved entities, relationships and/or events. In some embodiments, the database **160** includes an entity-relationship collection for multiple subscriptions. Subscriptions can be subscriptions of a particular tenant as described in FIG. **24**.

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In some embodiment, the graph projection manager **156** generates a graph projection for a particular user, application, subscription, and/or system. In this regard, the graph projection can be generated based on policies for the particular user, application, and/or system in addition to an ontology specific for that user, application, and/or system. In this regard, an entity could request a graph projection and the graph projection manager **156** can be configured to generate the graph projection for the entity based on policies and an ontology specific to the entity. The policies can indicate what entities, relationships, and/or events the entity has access to. The ontology can indicate what types of relationships between entities the requesting entity expects to see, e.g., floors within a building, devices within a floor, etc. Another requesting entity may have an ontology to see devices within a building and applications for the devices within the graph.

The graph projections generated by the graph projection manager **156** and stored in the graph projection database **162** can be a knowledge graph and is an integration point. For example, the graph projections can represent floor plans and systems associated with each floor. Furthermore, the graph projections can include events, e.g., telemetry data of the building subsystems **122**. The graph projections can show application services as nodes and API calls between the services as edges in the graph. The graph projections can illustrate the capabilities of spaces, users, and/or devices. The graph projections can include indications of the building subsystems **122**, e.g., thermostats, cameras, VAVs, etc. The graph projection database **162** can store graph projections that keep up a current state of a building.

The graph projections of the graph projection database **162** can be digital twins of a building. Digital twins can be digital replicas of physical entities that enable an in-depth analysis of data of the physical entities and provide the potential to monitor systems to mitigate risks, manage issues, and utilize simulations to test future solutions. Digital twins can play an important role in helping technicians find the root cause of issues and solve problems faster, in supporting safety and security protocols, and in supporting building managers in more efficient use of energy and other facilities resources. Digital twins can be used to enable and unify security systems, employee experience, facilities management, sustainability, etc.

In some embodiments the enrichment manager **138** can use a graph projection of the graph projection database **162** to enrich events. In some embodiments, the enrichment manager **138** can identify nodes and relationships that are associated with, and are pertinent to, the device that generated the event. For example, the enrichment manager **138** could identify a thermostat generating a temperature measurement event within the graph. The enrichment manager **138** can identify relationships between the thermostat and spaces, e.g., a zone that the thermostat is located in. The enrichment manager **138** can add an indication of the zone to the event.

Furthermore, the command processor **136** can be configured to utilize the graph projections to command the building subsystems **122**. The command processor **136** can identify a policy for a commanding entity within the graph projection to determine whether the commanding entity has the ability to make the command. For example, the command processor **136**, before allowing a user to make a command, determine, based on the graph projection database **162**, to determine that the user has a policy to be able to make the command.

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In some embodiments, the policies can be conditional based policies. For example, the building data platform **100** can apply one or more conditional rules to determine whether a particular system has the ability to perform an action. In some embodiments, the rules analyze a behavioral based biometric. For example, a behavioral based biometric can indicate normal behavior and/or normal behavior rules for a system. In some embodiments, when the building data platform **100** determines, based on the one or more conditional rules, that an action requested by a system does not match a normal behavior, the building data platform **100** can deny the system the ability to perform the action and/or request approval from a higher level system.

For example, a behavior rule could indicate that a user has access to log into a system with a particular IP address between 8 A.M. through 5 P.M. However, if the user logs in to the system at 7 P.M., the building data platform **100** may contact an administrator to determine whether to give the user permission to log in.

The change feed generator **152** can be configured to generate a feed of events that indicate changes to the digital twin, e.g., to the graph. The change feed generator **152** can track changes to the entities, relationships, and/or events of the graph. For example, the change feed generator **152** can detect an addition, deletion, and/or modification of a node or edge of the graph, e.g., changing the entities, relationships, and/or events within the database **160**. In response to detecting a change to the graph, the change feed generator **152** can generate an event summarizing the change. The event can indicate what nodes and/or edges have changed and how the nodes and edges have changed. The events can be posted to a topic by the change feed generator **152**.

The change feed generator **152** can implement a change feed of a knowledge graph. The building data platform **100** can implement a subscription to changes in the knowledge graph. When the change feed generator **152** posts events in the change feed, subscribing systems or applications can receive the change feed event. By generating a record of all changes that have happened, a system can stage data in different ways, and then replay the data back in whatever order the system wishes. This can include running the changes sequentially one by one and/or by jumping from one major change to the next. For example, to generate a graph at a particular time, all change feed events up to the particular time can be used to construct the graph.

The change feed can track the changes in each node in the graph and the relationships related to them, in some embodiments. If a user wants to subscribe to these changes and the user has proper access, the user can simply submit a web API call to have sequential notifications of each change that happens in the graph. A user and/or system can replay the changes one by one to reinstitute the graph at any given time slice. Even though the messages are "thin" and only include notification of change and the reference "id/seq id," the change feed can keep a copy of every state of each node and/or relationship so that a user and/or system can retrieve those past states at any time for each node. Furthermore, a consumer of the change feed could also create dynamic "views" allowing different "snapshots" in time of what the graph looks like from a particular context. While the twin manager **108** may contain the history and the current state of the graph based upon schema evaluation, a consumer can retain a copy of that data, and thereby create dynamic views using the change feed.

The schema and ontology **154** can define the message schema and graph ontology of the twin manager **108**. The message schema can define what format messages received

by the messaging manager **140** should have, e.g., what parameters, what formats, etc. The ontology can define graph projections, e.g., the ontology that a user wishes to view. For example, various systems, applications, and/or users can be associated with a graph ontology. Accordingly, when the graph projection manager **156** generates an graph projection for a user, system, or subscription, the graph projection manager **156** can generate a graph projection according to the ontology specific to the user. For example, the ontology can define what types of entities are related in what order in a graph, for example, for the ontology for a subscription of "Customer A," the graph projection manager **156** can create relationships for a graph projection based on the rule:

Region ↔ Building ↔ Floor ↔ Space ↔ Asset

For the ontology of a subscription of "Customer B," the graph projection manager **156** can create relationships based on the rule:

Building ↔ Floor ↔ Asset

The policy manager **158** can be configured to respond to requests from other applications and/or systems for policies. The policy manager **158** can consult a graph projection to determine what permissions different applications, users, and/or devices have. The graph projection can indicate various permissions that different types of entities have and the policy manager **158** can search the graph projection to identify the permissions of a particular entity. The policy manager **158** can facilitate fine grain access control with user permissions. The policy manager **158** can apply permissions across a graph, e.g., if "user can view all data associated with floor 1" then they see all subsystem data for that floor, e.g., surveillance cameras, HVAC devices, fire detection and response devices, etc.

The twin manager **108** includes a query manager **165** and a twin function manager **167**. The query manager **165** can be configured to handle queries received from a requesting system, e.g., the user device **176**, the applications **110**, and/or any other system. The query manager **165** can receive queries that include query parameters and context. The query manager **165** can query the graph projection database **162** with the query parameters to retrieve a result. The query manager **165** can then cause an event processor, e.g., a twin function, to operate based on the result and the context. In some embodiments, the query manager **165** can select the twin function based on the context and/or perform operations based on the context. In some embodiments, the query manager **165** is configured to perform the operations described with reference to FIGS. **5-10**.

The twin function manager **167** can be configured to manage the execution of twin functions. The twin function manager **167** can receive an indication of a context query that identifies a particular data element and/or pattern in the graph projection database **162**. Responsive to the particular data element and/or pattern occurring in the graph projection database **162** (e.g., based on a new data event added to the graph projection database **162** and/or change to nodes or edges of the graph projection database **162**, the twin function manager **167** can cause a particular twin function to execute. The twin function can execute based on an event, context, and/or rules. The event can be data that the twin function executes against. The context can be information that provides a contextual description of the data, e.g., what device the event is associated with, what control point should be

updated based on the event, etc. The twin function manager **167** can be configured to perform the operations of the FIGS. **11-15**.

Referring now to FIG. **2**, a graph projection **200** of the twin manager **108** including application programming interface (API) data, capability data, policy data, and services is shown, according to an exemplary embodiment. The graph projection **200** includes nodes **202-240** and edges **250-272**. The nodes **202-240** and the edges **250-272** are defined according to the key **201**. The nodes **202-240** represent different types of entities, devices, locations, points, persons, policies, and software services (e.g., API services). The edges **250-272** represent relationships between the nodes **202-240**, e.g., dependent calls, API calls, inferred relationships, and schema relationships (e.g., BRICK relationships).

The graph projection **200** includes a device hub **202** which may represent a software service that facilitates the communication of data and commands between the cloud platform **106** and a device of the building subsystems **122**, e.g., door actuator **214**. The device hub **202** is related to a connector **204**, an external system **206**, and a digital asset "Door Actuator" **208** by edge **250**, edge **252**, and edge **254**.

The cloud platform **106** can be configured to identify the device hub **202**, the connector **204**, the external system **206** related to the door actuator **214** by searching the graph projection **200** and identifying the edges **250-254** and edge **258**. The graph projection **200** includes a digital representation of the "Door Actuator," node **208**. The digital asset "Door Actuator" **208** includes a "DeviceNameSpace" represented by node **207** and related to the digital asset "Door Actuator" **208** by the "Property of Object" edge **256**.

The "Door Actuator" **214** has points and timeseries. The "Door Actuator" **214** is related to "Point A" **216** by a "has_a" edge **260**. The "Door Actuator" **214** is related to "Point B" **218** by a "has_A" edge **258**. Furthermore, timeseries associated with the points A and B are represented by nodes "TS" **220** and "TS" **222**. The timeseries are related to the points A and B by "has_a" edge **264** and "has_a" edge **262**. The timeseries "TS" **220** has particular samples, sample **210** and **212** each related to "TS" **220** with edges **268** and **266** respectively. Each sample includes a time and a value. Each sample may be an event received from the door actuator that the cloud platform **106** ingests into the entity, relationship, and event database **160**, e.g., ingests into the graph projection **200**.

The graph projection **200** includes a building **234** representing a physical building. The building includes a floor represented by floor **232** related to the building **234** by the "has_a" edge from the building **234** to the floor **232**. The floor has a space indicated by the edge "has_a" **270** between the floor **232** and the space **230**. The space has particular capabilities, e.g., is a room that can be booked for a meeting, conference, private study time, etc. Furthermore, the booking can be canceled. The capabilities for the floor **232** are represented by capabilities **228** related to space **230** by edge **280**. The capabilities **228** are related to two different commands, command "book room" **224** and command "cancel booking" **226** related to capabilities **228** by edge **284** and edge **282** respectively.

If the cloud platform **106** receives a command to book the space represented by the node, space **230**, the cloud platform **106** can search the graph projection **200** for the capabilities for the **228** related to the space **230** to determine whether the cloud platform **106** can book the room.

In some embodiments, the cloud platform **106** could receive a request to book a room in a particular building, e.g., the building **234**. The cloud platform **106** could search

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the graph projection 200 to identify spaces that have the capabilities to be booked, e.g., identify the space 230 based on the capabilities 228 related to the space 230. The cloud platform 106 can reply to the request with an indication of the space and allow the requesting entity to book the space 230.

The graph projection 200 includes a policy 236 for the floor 232. The policy 236 is related set for the floor 232 based on a “To Floor” edge 274 between the policy 236 and the floor 232. The policy 236 is related to different roles for the floor 232, read events 238 via edge 276 and send command 240 via edge 278. The policy 236 is set for the entity 203 based on has edge 251 between the entity 203 and the policy 236.

The twin manager 108 can identify policies for particular entities, e.g., users, software applications, systems, devices, etc. based on the policy 236. For example, if the cloud platform 106 receives a command to book the space 230. The cloud platform 106 can communicate with the twin manager 108 to verify that the entity requesting to book the space 230 has a policy to book the space. The twin manager 108 can identify the entity requesting to book the space as the entity 203 by searching the graph projection 200. Furthermore, the twin manager 108 can further identify the edge has 251 between the entity 203 and the policy 236 and the edge 1178 between the policy 236 and the command 240.

Furthermore, the twin manager 108 can identify that the entity 203 has the ability to command the space 230 based on the edge 1174 between the policy 236 and the edge 270 between the floor 232 and the space 230. In response to identifying the entity 203 has the ability to book the space 230, the twin manager 108 can provide an indication to the cloud platform 106.

Furthermore, if the entity makes a request to read events for the space 230, e.g., the sample 210 and the sample 212, the twin manager 108 can identify the edge has 251 between the entity 203 and the policy 236, the edge 1178 between the policy 236 and the read events 238, the edge 1174 between the policy 236 and the floor 232, the “has_a” edge 270 between the floor 232 and the space 230, the edge 268 between the space 230 and the door actuator 214, the edge 260 between the door actuator 214 and the point A 216, the “has_a” edge 264 between the point A 216 and the TS 220, and the edges 268 and 266 between the TS 220 and the samples 210 and 212 respectively.

Referring now to FIG. 3, a graph projection 300 of the twin manager 108 including application programming interface (API) data, capability data, policy data, and services is shown, according to an exemplary embodiment. The graph projection 300 includes the nodes and edges described in the graph projection 200 of FIG. 2. The graph projection 300 includes a connection broker 354 related to capabilities 228 by edge 398a. The connection broker 354 can be a node representing a software application configured to facilitate a connection with another software application. In some embodiments, the cloud platform 106 can identify the system that implements the capabilities 228 by identifying the edge 398a between the capabilities 228 and the connection broker 354.

The connection broker 354 is related to an agent that optimizes a space 356 via edge 398b. The agent represented by the node 356 can book and cancel bookings for the space represented by the node 230 based on the edge 398b between the connection broker 354 and the node 356 and the edge 398a between the capabilities 228 and the connection broker 354.

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The connection broker 354 is related to a cluster 308 by edge 398c. Cluster 308 is related to connector B 302 via edge 398e and connector A 306 via edge 398d. The connector A 306 is related to an external subscription service 304. A connection broker 310 is related to cluster 308 via an edge 311 representing a rest call that the connection broker represented by node 310 can make to the cluster represented by cluster 308.

The connection broker 310 is related to a virtual meeting platform 312 by an edge 354. The node 312 represents an external system that represents a virtual meeting platform. The connection broker represented by node 310 can represent a software component that facilitates a connection between the cloud platform 106 and the virtual meeting platform represented by node 312. When the cloud platform 106 needs to communicate with the virtual meeting platform represented by the node 312, the cloud platform 106 can identify the edge 354 between the connection broker 310 and the virtual meeting platform 312 and select the connection broker represented by the node 310 to facilitate communication with the virtual meeting platform represented by the node 312.

A capabilities node 318 can be connected to the connection broker 310 via edge 360. The capabilities 318 can be capabilities of the virtual meeting platform represented by the node 312 and can be related to the node 312 through the edge 360 to the connection broker 310 and the edge 354 between the connection broker 310 and the node 312. The capabilities 318 can define capabilities of the virtual meeting platform represented by the node 312. The node 320 is related to capabilities 318 via edge 362. The capabilities may be an invite bob command represented by node 316 and an email bob command represented by node 314. The capabilities 318 can be linked to a node 320 representing a user, Bob. The cloud platform 106 can facilitate email commands to send emails to the user Bob via the email service represented by the node 304. The node 304 is related to the connect a node 306 via edge 398f. Furthermore, the cloud platform 106 can facilitate sending an invite for a virtual meeting via the virtual meeting platform represented by the node 312 linked to the node 318 via the edge 358.

The node 320 for the user Bob can be associated with the policy 236 via the “has” edge 364. Furthermore, the node 320 can have a “check policy” edge 366 with a portal node 324. The device API node 328 has a check policy edge 370 to the policy node 236. The portal node 324 has an edge 368 to the policy node 236. The portal node 324 has an edge 323 to a node 326 representing a user input manager (UIM). The portal node 324 is related to the UIM node 326 via an edge 323. The UIM node 326 has an edge 323 to a device API node 328. The UIM node 326 is related to the door actuator node 214 via edge 372. The door actuator node 214 has an edge 374 to the device API node 328. The door actuator 214 has an edge 335 to the connector virtual object 334. The device hub 332 is related to the connector virtual object via edge 380. The device API node 328 can be an API for the door actuator 214. The connector virtual object 334 is related to the device API node 328 via the edge 331.

The device API node 328 is related to a transport connection broker 330 via an edge 329. The transport connection broker 330 is related to a device hub 332 via an edge 378. The device hub represented by node 332 can be a software component that hands the communication of data and commands for the door actuator 214. The cloud platform 106 can identify where to store data within the graph projection 300 received from the door actuator by identifying the nodes and edges between the points 216 and 218 and

the device hub node 332. Similarly, the cloud platform 308 can identify commands for the door actuator that can be facilitated by the device hub represented by the node 332, e.g., by identifying edges between the device hub node 332 and an open door node 352 and an lock door node 350. The door actuator 114 has an edge “has mapped an asset” 280 between the node 214 and a capabilities node 348. The capabilities node 348 and the nodes 352 and 350 are linked by edges 396 and 394.

The device hub 332 is linked to a cluster 336 via an edge 384. The cluster 336 is linked to connector A 340 and connector B 338 by edges 386 and the edge 389. The connector A 340 and the connector B 338 is linked to an external system 344 via edges 388 and 390. The external system 344 is linked to a door actuator 342 via an edge 392.

Referring now to FIG. 4, a graph projection 400 of the twin manager 108 including equipment and capability data for the equipment is shown, according to an exemplary embodiment. The graph projection 400 includes nodes 402-456 and edges 360-498f. The cloud platform 106 can search the graph projection 400 to identify capabilities of different pieces of equipment.

A building node 404 represents a particular building that includes two floors. A floor 1 node 402 is linked to the building node 404 via edge 460 while a floor 2 node 406 is linked to the building node 404 via edge 462. The floor 2 includes a particular room 2023 represented by edge 464 between floor 2 node 406 and room 2023 node 408. Various pieces of equipment are included within the room 2023. A light represented by light node 416, a bedside lamp node 414, a bedside lamp node 412, and a hallway light node 410 are related to room 2023 node 408 via edge 466, edge 472, edge 470, and edge 468.

The light represented by light node 416 is related to a light connector 426 via edge 484. The light connector 426 is related to multiple commands for the light represented by the light node 416 via edges 484, 486, and 488. The commands may be a brightness setpoint 424, an on command 425, and a hue setpoint 428. The cloud platform 106 can receive a request to identify commands for the light represented by the light 416 and can identify the nodes 424-428 and provide an indication of the commands represented by the node 424-428 to the requesting entity. The requesting entity can then send commands for the commands represented by the nodes 424-428.

The bedside lamp node 414 is linked to a bedside lamp connector 481 via an edge 413. The connector 481 is related to commands for the bedside lamp represented by the bedside lamp node 414 via edges 492, 496, and 494. The command nodes are a brightness setpoint node 432, an on command node 434, and a color command 436. The hallway light 410 is related to a hallway light connector 446 via an edge 498d. The hallway light connector 446 is linked to multiple commands for the hallway light node 410 via edges 498g, 498f, and 498e. The commands are represented by an on command node 452, a hue setpoint node 450, and a light bulb activity node 448.

The graph projection 400 includes a name space node 422 related to a server A node 418 and a server B node 420 via edges 474 and 476. The name space node 422 is related to the bedside lamp connector 481, the bedside lamp connector 444, and the hallway light connector 446 via edges 482, 480, and 478. The bedside lamp connector 444 is related to commands, e.g., the color command node 440, the hue setpoint command 438, a brightness setpoint command 456, and an on command 454 via edges 498c, 498b, 498a, and 498.

Referring now to FIG. 5, a system 500 for managing a digital twin where an artificial intelligence agent can be executed to infer and/or predict information for an entity of a graph is shown, according to an exemplary embodiment. The system 500 can be components of the building data platform 100, e.g., components run on the processors and memories of the edge platform 102, the cloud platform 106, the twin manager 108, and/or the applications 110. The system 500 can, in some implementations, implement a digital twin with artificial intelligence.

A digital twin (or a shadow) may be a computing entity that describes a physical thing (e.g., a building, spaces of a building, devices of a building, people of the building, equipment of the building, etc.) through modeling the physical thing through a set of attributes that define the physical thing. A digital twin can refer to a digital replica of physical assets (a physical device twin) and can be extended to store processes, people, places, systems that can be used for various purposes. The digital twin can include both the ingestion of information and actions learned and executed through artificial intelligence agents.

In FIG. 5, the digital twin can be a graph 529 managed by the twin manager 108 and/or artificial intelligence agents 570. In some embodiments, the digital twin is the combination of the graph 529 with the artificial intelligence agents 570. In some embodiments, the digital twin enables the creation of a chronological time-series database of telemetry events for analytical purposes. In some embodiments, the graph 529 uses the BRICK schema.

The twin manager 108 stores the graph 529 which may be a graph data structure including various nodes and edges interrelating the nodes. The graph 529 may be the same as, or similar to, the graph projections described herein with reference to FIGS. 1-4. The graph 529 includes nodes 510-526 and edges 528-546. The graph 529 includes a building node 526 representing a building that has a floor indicated by the “has” edge 546 to the floor node 522. The floor node 522 is related to a zone node 510 via a “has” edge 544 indicating that the floor represented by the node 522 has a zone represented by the zone 510.

The floor node 522 is related to the zone node 518 by the “has” edge 540 indicating that the floor represented by the floor node 522 has another zone represented by the zone node 518. The floor node 522 is related to another zone node 524 via a “has” edge 542 representing that the floor represented by the floor node 522 has a third zone represented by the zone node 524.

The graph 529 includes an AHU node 514 representing an AHU of the building represented by the building node 526. The AHU node 514 is related by a “supplies” edge 530 to the VAV node 512 to represent that the AHU represented by the AHU node 514 supplies air to the VAV represented by the VAV node 512. The AHU node 514 is related by a “supplies” edge 536 to the VAV node 520 to represent that the AHU represented by the AHU node 514 supplies air to the VAV represented by the VAV node 520. The AHU node 514 is related by a “supplies” edge 532 to the VAV node 516 to represent that the AHU represented by the AHU node 514 supplies air to the VAV represented by the VAV node 516.

The VAV node 516 is related to the zone node 518 via the “serves” edge 534 to represent that the VAV represented by the VAV node 516 serves (e.g., heats or cools) the zone represented by the zone node 518. The VAV node 520 is related to the zone node 524 via the “serves” edge 538 to represent that the VAV represented by the VAV node 520 serves (e.g., heats or cools) the zone represented by the zone node 524. The VAV node 512 is related to the zone node 510

via the “serves” edge **528** to represent that the VAV represented by the VAV node **512** serves (e.g., heats or cools) the zone represented by the zone node **510**.

Furthermore, the graph **529** includes an edge **533** related to a timeseries node **564**. The timeseries node **564** can be information stored within the graph **529** and/or can be information stored outside the graph **529** in a different database (e.g., a timeseries database). In some embodiments, the timeseries node **564** stores timeseries data (or any other type of data) for a data point of the VAV represented by the VAV node **516**. The data of the timeseries node **564** can be aggregated and/or collected telemetry data of the timeseries node **564**.

Furthermore, the graph **529** includes an edge **537** related to a timeseries node **566**. The timeseries node **566** can be information stored within the graph **529** and/or can be information stored outside the graph **529** in a different database (e.g., a timeseries database). In some embodiments, the timeseries node **566** stores timeseries data (or any other type of data) for a data point of the VAV represented by the VAV node **516**. The data of the timeseries node **564** can be inferred information, e.g., data inferred by one of the artificial intelligence agents **570** and written into the timeseries node **564** by the artificial intelligence agent **570**. In some embodiments, the timeseries **546** and/or **566** are stored in the graph **529** but are stored as references to timeseries data stored in a timeseries database.

The twin manager **108** includes various software components. For example, the twin manager **108** includes a device management component **548** for managing devices of a building. The twin manager **108** includes a tenant management component **550** for managing various tenant subscriptions. The twin manager **108** includes an event routing component **552** for routing various events. The twin manager **108** includes an authentication and access component **554** for performing user and/or system authentication and granting the user and/or system access to various spaces, pieces of software, devices, etc. The twin manager **108** includes a commanding component **556** allowing a software application and/or user to send commands to physical devices. The twin manager **108** includes an entitlement component **558** that analyzes the entitlements of a user and/or system and grants the user and/or system abilities based on the entitlements. The twin manager **108** includes a telemetry component **560** that can receive telemetry data from physical systems and/or devices and ingest the telemetry data into the graph **529**. Furthermore, the twin manager **108** includes an integrations component **562** allowing the twin manager **108** to integrate with other applications.

The twin manager **108** includes a gateway **506** and a twin connector **508**. The gateway **506** can be configured to integrate with other systems and the twin connector **508** can be configured to allow the gateway **506** to integrate with the twin manager **108**. The gateway **506** and/or the twin connector **508** can receive an entitlement request **502** and/or an inference request **504**. The entitlement request **502** can be a request received from a system and/or a user requesting that an AI agent action be taken by the AI agent **570**. The entitlement request **502** can be checked against entitlements for the system and/or user to verify that the action requested by the system and/or user is allowed for the user and/or system. The inference request **504** can be a request that the AI agent **570** generates an inference, e.g., a projection of information, a prediction of a future data measurement, an extrapolated data value, etc.

The cloud platform **106** is shown to receive an entitlement request **586**. The request **586** can be received from a system,

application, and/or user device (e.g., from the applications **110**, the building subsystems **122**, and/or the user device **176**). The entitlement request **586** may be a manual or automated request. In some embodiments, the request **586** can be automated via an API request, e.g., when such an entitlement to an AI model is enabled for a particular tenant/end-customer of the data platform **100**. In some embodiments, the request **586** may be a manual request by a person asking for an inference/future state of the building. The request **586** could be automatically made to keep the state updated on a particular cadence. In some embodiments, a request for a future state of one digital twin can cause a re-calculation of the future states of related digital twins automatically. For example, based on a request asking for the future occupancy of a room in a building in the coming week, a re-calculation of the future AHU operating value recommendations to reduce the spread of infectious diseases may be triggered. The results of these automatic triggers determining future states not directly requested by the request **586** can be collected and presented to the caller of the request **586** as causal future states.

The entitlement request **586** may be a request for the AI agent **570** to perform an action, e.g., an action that the requesting system and/or user has an entitlement for. The cloud platform **106** can receive the entitlement request **586** and check the entitlement request **586** against an entitlement database **584** storing a set of entitlements to verify that the requesting system has access to the user and/or system. The cloud platform **106**, responsive to the entitlement request **586** being approved, can create a job for the AI agent **570** to perform. The created job can be added to a job request topic **580** of a set of topics **578**.

The job request topic **580** can be fed to AI agents **570**. For example, the topics **580** can be fanned out to various AI agents **570** based on the AI agent that each of the topics **580** pertains to (e.g., based on an identifier that identifies an agent and is included in each job of the topic **580**). The AI agents **570** include a service client **572**, a connector **574**, and a model **576**. The model **576** can be loaded into the AI agent **570** from a set of AI models stored in the AI model storage **568**. The AI model storage **568** can store models for making energy load predictions for a building, weather forecasting models for predicting a weather forecast, action/decision models to take certain actions responsive to certain conditions being met, an occupancy model for predicting occupancy of a space and/or a building, etc. The models of the AI model storage **568** can be neural networks (e.g., convolutional neural networks, recurrent neural networks, deep learning networks, etc.), decision trees, support vector machines, and/or any other type of artificial intelligence, machine learning, and/or deep learning category. In some embodiments, the models are rule based triggers and actions that include various parameters for setting a condition and defining an action.

The AI agent **570** can include triggers **595** and actions **597**. The triggers **595** can be conditional rules that, when met, cause one or more of the actions **597**. The triggers **595** can be executed based on information stored in the graph **529** and/or data received from the building subsystems **122**. The actions **597** can be executed to determine commands, actions, and/or outputs. The output of the actions **597** can be stored in the graph **529** and/or communicated to the building subsystems **122**.

In some embodiments, the triggers **595** and/or the actions **597** are stored as nodes in the graph **529**. The nodes for the triggers **595** and/or the actions **597** can be stored as capability nodes. Capability nodes may be nodes that describe

operations that a digital twin can perform (e.g., is configured and/or capable of performing). Capabilities are described with reference to FIGS. 1-4. In some embodiments, the graph 529 stores a first node representing a particular digital twin, agent, artificial intelligence, machine learning entity, etc. The graph 529 can further store one or more second nodes related to the first node via one or more edges (or one or more nodes). The one or more second nodes may describe capabilities of the digital twin, agent, artificial intelligence, machine learning entity, etc. The capabilities may be triggers and/or actions (e.g., the triggers 595 and/or actions 597). The first node may be related to a particular component node (e.g., the VAV node 516) via another edge to represent that the digital twin, agent, artificial intelligence, machine learning entity, etc. makes inferences or other data determinations for the component (e.g., a VAV) represented by the particular component node (e.g., the VAV node 516).

The AI agent 570 can include a service client 572 that causes an instance of an AI agent to run. The instance can be hosted by the artificial intelligence service client 588. The client 588 can cause a client instance 592 to run and communicate with the AI agent 570 via a gateway 590. The client instance 592 can include a service application 594 that interfaces with a core algorithm 598 via a functional interface 596. The core algorithm 598 can run the model 576, e.g., train the model 576 and/or use the model 576 to make inferences and/or predictions.

In some embodiments, the core algorithm 598 can be configured to perform learning based on the graph 529. In some embodiments, the core algorithm 598 can read and/or analyze the nodes and relationships of the graph 529 to make decisions. In some embodiments, the core algorithm 598 can be configured to use telemetry data (e.g., the timeseries data 564) from the graph 529 to make inferences on and/or perform model learning. In some embodiments, the result of the inferences can be the timeseries 566. In some embodiments, the timeseries 564 is an input into the model 576 that predicts the timeseries 566.

In some embodiments, the core algorithm 598 can generate the timeseries 566 as an inference for a data point, e.g., a prediction of values for the data point at future times. The timeseries 564 may be actual data for the data point. In this regard, the core algorithm 598 can learn and train by comparing the inferred data values against the true data values. In this regard, the model 576 can be trained by the core algorithm 598 to improve the inferences made by the model 576.

Referring now to FIG. 6, a process 600 for executing an artificial intelligence agent to infer and/or predict information is shown, according to an exemplary embodiment. The process 600 can be performed by the system 500 and/or components of the system 500. The process 600 can be performed by the building data platform 100. Furthermore, the process 600 can be performed by any computing device described herein.

In step 602, the twin manager 108 receives information from a physical device and stores the information, or a link to the information, in the graph 529. For example, the telemetry component 560 can receive telemetry data from physical devices, e.g., the building subsystems 122. The telemetry can be measured data values, a log of historical equipment commands, etc. The telemetry component 560 can store the received information in the graph 529 by relating a node storing the information to a node representing the physical device. For example, the telemetry component 560 can store timeseries data as the timeseries 566 along by identifying that the physical device is a VAV

represented by the VAV node 516 and that an edge 537 relates the VAV node 516 to the timeseries node 566.

In step 604, the twin manager 108 and/or the cloud platform 106 receives an indication to execute an artificial intelligence agent of an entity represented in the graph 529, the AI agent being associated with a model. In some embodiments, the indication is created by a user and provided via the user device 176. In some embodiments, the indication is created by an application, e.g., one of the applications 110. In some embodiments, the indication is a triggering event that triggers the agent and is received from the building subsystems 122 and/or another agent (e.g., an output of one agent fed into another agent).

In some embodiments, the AI agent is an agent for a specific entity represented in the graph 529. For example, the agent could be a VAV maintenance agent configured to identify whether a VAV (e.g., a VAV represented by the nodes 512, 530, and/or 516) should have maintenance performed at a specific time. Another agent could be a floor occupant prediction agent that is configured to predict the occupancy of a particular floor of a building, e.g., the floor represented by the floor node 522.

Responsive to receiving the indication, in step 606, the AI agent 570 causes a client instance 592 to run the model 576 based on the information received in step 602. In some embodiments, the information received in step 602 is provided directly to the AI agent 570. In some embodiments, the information is read from the graph 529 by the AI agent 570.

In step 608, the AI agent 570 stores the inferred and/or predicted information in the graph 529 (or stores the inferred and/or predicted information in a separate data structure with a link to the graph 529). In some embodiments, the AI agent 570 identifies that the node that represents the physical entity that the AI agent 570 inferred and/or predicted information for, e.g., the VAV represented by the VAV 516. The AI agent 570 can identify that the timeseries node 566 stores the inferred and/or predicted information by identifying the edge 537 between the VAV node 516 and the timeseries node 566.

In step 610, the AI agent 570 can retrieve the inferred or predicted information from the graph 529 responsive to receiving an indication to execute the model of the AI agent 570 of the inferred or predicted information, e.g., similar to the step 604. In step 612, the AI agent 570 can execute one or more actions based on the inferred and/or predicted information of the step 610 based the inferred and/or predicted information retrieved from the graph 529. In some embodiments, the AI agent 570 executes the model 576 based on the inferred and/or predicted information.

In step 614, the AI agent 570 can train the model 576 based on the inferred or predicted information read from the graph 529 and received actual values for the inferred or predicted information. In some embodiments, the AI agent 570 can train and update parameters of the model 576. For example, the timeseries 564 may represent actual values for a data point of the VAV represented by the VAV node 516. The timeseries 566 can be the inferred and/or predicted information. The AI agent 570 can compare the timeseries 564 and the timeseries 566 to determine an error in the inferences and/or predictions of the model 576. The error can be used by the model 576 to update and train the model 576.

Referring now to FIG. 7, a digital twin 700 including a connector and a database is shown, according to an exemplary embodiment. The digital twin 700 can be a software component stored and/or managed by the building data platform 100. The building data platform 100 includes

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connectors **702** and a database **704**. The database **704** can store data attributes for a physical entity, e.g., a building, a VAV, etc. that describe the current state and/or operation of the physical entity. The connector **702** can be a software component that receives data from the physical device represented by the digital twin **700** and updates the attributes of the database **704**. For example, the connector **702** can ingest device telemetry data into the database **704** to update the attributes of the digital twin **700**.

Referring now to FIG. 8, a digital twin **800** including triggers **802**, connectors **804**, actions **806**, and a graph **808** is shown, according to an exemplary embodiment. The digital twin **800** can be a digital replica of physical assets (e.g., a physical device twin, sensor twin, actuator twin, building device twin, etc.) and can be used to store processes, people, places, systems that can be used for various purposes. The digital twins can be created, managed, stored, and/or operated on by the building data platform **100**.

In some cases, the devices can also be actuated on (told to perform an action). For example, a thermostat has sensors to measure temperature and humidity. A thermostat can also be asked to perform an action of setting the setpoint for a HVAC system. In this regard, the digital twin **800** can be configured so that information that the digital twin **800** can be made aware of can be stored by the digital twin **800** and there are also actions that the digital twin **800** can take.

The digital twin **800** can include a connector **804** that ingests device telemetry into the graph **808** and/or update the digital twin attributes stored in the graph **808**. In some embodiments, the connectors **804** can ingest external data received from external data sources into the graph **808**. The external data could be weather data, calendar data, etc. In some embodiments, the connectors **804** can send commands back to the devices, e.g., the actions determined by the actions **806**.

The digital twin **800** includes triggers **802** which can set conditional logic for triggering the actions **706**. The digital twin **800** can apply the attributes stored in the graph **808** against a rule of the triggers **802**. When a particular condition of the rule of the triggers **802** involving that attribute is met, the actions **706** can execute. One example of a trigger could be a conditional question, “when the temperature of the zone managed by the thermostat reaches x degrees Fahrenheit.” When the question is met by the attributes stored in the graph **808**, a rule of the actions **706** can execute.

The digital twin **800** can, when executing the actions **806**, update an attribute of the graph **808**, e.g., a setpoint, an operating setting, etc. These attributes can be translated into commands that the building data platform **100** can send to physical devices that operate based on the setpoint, the operating setting, etc. An example of an action rule for the actions **806** could be the statement, “update the setpoint of the HVAC system for a zone to x Degrees Fahrenheit.”

In some embodiments, the triggers **802** and/or the actions **806** are predefined and/or manually defined through user input of the user device **176**. In some cases, it may be difficult for a user to determine what the parameter values of the trigger rule should be (e.g., what values maximize a particular reward or minimize a particular penalty). Similarly, it may be difficult for a user to determine what the parameter values of the action rule should be (e.g., what values maximize the particular reward or minimize the particular penalty). Furthermore, even if the user is able to identify the ideal parameter values for the triggers **802** and the actions **806**, the ideal values for the parameters may not be constant and may instead change over time. Therefore, it would be desirable if the values of the attributes for the

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triggers **802** and the actions **806** are tuned optimally and automatically by the building data platform **100** by observing the responses from other related digital twins.

Causal patterns between one or more digital twins having their triggering conditions satisfied and one or more digital twins (including the triggering digital twin) actuating by sending specific commands to their physical counterparts could be learned and defined by the building data platform **100**. Automated learning can be used by the building data platform **100** during real operations, by running simulations using digital twins, or using predicted inference within the digital twin. There may not even be the need for all standard operating procedures in building systems to be defined upfront by a user since patterns of interaction between digital twins can be learned by the building data platform **100** to define and recommend those to building and facility owners.

Referring now to FIG. 9, a system **900** of digital twins including a people counter digital twin **902**, an HVAC digital twin **904**, and a facility manager digital twin **906** that are interconnected is shown, according to an exemplary embodiment. In FIG. 9, the people counter digital twin **902** includes triggers **908**, connectors **910**, actions, and/or the graph **926**.

The system **900** further includes an HVAC digital twin **904** that includes triggers, connectors **916**, actions **918**, and/or the graph **928**. The system further includes the facility manager **906** that includes triggers, connectors **922**, actions **924**, and/or the graph **930**. In some embodiments, the graph **926**, the graph **928**, and the graph **930** are the same graph or different graphs. In some embodiments, the graphs **926-930** are the graph **529**.

The people counter digital twin **902** can output a “low occupancy” attribute which can be stored in the graph **926** and/or provided to the HVAC digital twin **904** and/or the facility manager digital twin **906**, in some embodiments. In some embodiments, if all of the digital twins use and/or have access to the same graph, if the people counter digital twin **902** stores the low occupancy indicator in the graph, the HVAC digital twin **904** and the facility manager digital twin **906** can read the attribute from the graph. In some embodiment, a rules engine **926** can store rules that link the twins **902**, **904**, and **906** together. The rules engine **926** can determine whether the triggers of the various digital twins are triggered, e.g., the trigger **908**. Furthermore, the rules engine **926** can determine (based on one or more rules stored by the rules engine **926**) that if the trigger **908** of the people counter digital twin **902** is triggered, the actions **918** of the HVAC digital twin **904** and the actions **924** of the facility manager digital twin **906** should be executed.

In some embodiments, the trigger **908** is the logical condition, “when there are less than twenty people in a particular area.” Responsive to an occupancy count of the particular area is less than twenty, which the people counter digital twin **902** can determine from models and/or information of the graph **926**, a low occupancy indication can be generated.

Responsive to the trigger **908** being triggered, the rules engine **926** can cause the actions **918** to execute to switch an HVAC mode to an economy mode. The economy mode status for an HVAC system can be stored in the graph **928** and/or communicated to an HVAC controller to execute on. Responsive to the trigger **908** being triggered, the rules engine **926** can cause the actions **924** to execute to notify a facility manager of the low occupancy status, e.g., send a notification to a user device of the facility manager.

In some embodiments, the digital twins of the system **900** can be solution twins, e.g., the people counter twin **902**, the HVAC digital twin **904**, the facility manager twin **906**, etc. The digital twin can be a solution twin because it represents a particular software solutions for the building. For example, in some embodiments, an occupancy sensor digital twin of a zone could be triggered with under-utilized criteria (e.g., the triggering of the people counter digital twin **902** shown in FIG. **9**). The people counter digital twin **902** could be configured to identify what AHU is serving the zone that it has made an occupancy detection for based on the nodes and/or edges of the graph **926** relating a zone node for the zone and an AHU node for the AHU. In some embodiments, the AHU digital twin can evaluate the desired setting for the zone through running a simulation with one or more models. In some embodiments, an FM digital twin can evaluate space arrangement and/or purposing.

Referring now to FIG. **10**, a system **1000** including an employee digital twin **1002**, a calendar digital twin **1006**, a meeting room digital twin **1004**, and a cafeteria digital twin **1008** that are interconnected is shown, according to an exemplary embodiment. The system **1000** includes a solution digital twin for an employee, a meeting room, a cafeteria, and a calendar. In the system **1000**, an employee digital twin **1002** and a calendar digital twin **1006** trigger and a rules engine **926** causes one or more associated digital twins, a meeting room digital twin **1004** and a cafeteria digital twin **1008** to execute responsive to the triggers. The calendar digital twin **1006** can include a connector **1016**, the meeting room digital twin can include a connector **1022**, and the cafeteria digital twin **1008** can include a connector **1028** for ingesting information into the graphs **1034-1038**. In some embodiment, the rules engine **926** can store rules that link the twins **1002**, **1006**, **1004**, and/or **1008** together. The rules engine **926** can determine whether the triggers of the various digital twins are triggered, e.g., the triggers **1010** and/or **1014**. Furthermore, the rules engine **926** can determine (based on one or more rules stored by the rules engine **926**) that if the triggers **1010** and **1014** are triggered, the actions **1024** and **1030** should be executed.

In FIG. **10**, the employee digital twin **1002** includes a graph **1032**, the calendar digital twin **1006** includes a graph **1036**, the meeting room digital twin **1004** includes a graph **1034**, and the cafeteria digital twin **1008** includes a graph **1038**. The graphs **1032-1038** can be the same graphs and/or different graphs and can be the same as, or similar to, the graph **529**.

The employee digital twin **1002** can generate an indication of whether a particular occupant is near a particular office via an action responsive to the trigger **1010** triggering when a particular occupant is a particular instance (e.g., 250 meters) from their office. The digital twin **1002** can identify the occupant, the occupant's office, and the location of the office through analyzing the nodes and/or edge of the graph **1032**. The calendar digital twin **1006** determines, based on calendar data (e.g., calendar data stored in the graph **1036**), whether it is a work day via the trigger **1014** (e.g., is a day Monday through Friday). Responsive to determining that it is a work day, the calendar digital twin **1006** generates an indication that it is a work day via actions of the calendar digital twin **1006**.

The meeting room digital twin **1004** can receive the work day indication from the calendar digital twin **1006** and can receive the occupant near office indication from the employee digital twin **1002**. The rules engine **926** can cause the meeting room digital twin **1004** to take actions to reserve a meeting room via the actions **1024** responsive to the

triggers **1010** and **1014** being triggered. The rules engine **926** can cause the cafeteria digital twin **1008** to trigger the ordering of a coffee for the occupant via the action **1030** responsive to the triggers **1010** and **1014** being triggered.

Referring now to FIG. **11**, a process **1100** of an agent executing a trigger rule and an action rule is shown, according to an exemplary embodiment. The process **1100** can be performed by the system **500** and/or components of the system **500**. In some embodiments, the building data platform **100** can perform the process **1100**. Furthermore, the process **1100** can be performed by any computing device described herein.

In step **1102**, the building data platform can store an agent **570** in a data structure. The agent **570** can include a trigger rule indicating a condition for executing an action rule and an action rule indicating an action to be performed responsive to the condition being met. In some embodiments, the model **576** includes, or can be replaced with, the trigger rule and the action rule. The trigger rule and the action rule can be logical statements and/or conditions that include parameter values and/or create an output action. The parameter values can, in some embodiments, be identified through a learning process, e.g., as described through FIGS. **12-22**.

In step **1104**, the agent **570** can receive information from at least one of a physical device and/or from the graph **529**. The information can be generated by a physical device, e.g., the building subsystems **122**. The building data platform **100** can, in some embodiments, receive the information from the physical device, ingest the information into the graph **529**, and the agent **570** can read the information from the graph **529**. In some embodiments, the agent **570** can check the information of the graph **529** against a trigger rule at a set period.

In step **1106**, the agent **570** determines whether the information received in the step **1104** causes the condition to be met. The agent **570** can apply the information to the trigger rule to determine whether the trigger rule is triggered, i.e., the condition of the trigger rule being met.

In step **1108**, the agent **570** can perform the action responsive to the condition being met by the information determined in step **1106**. The action may cause a physical device to be operated or information be sent to another agent including another trigger rule and another action rule. In some embodiments, the action can be performed by executing the action rule of the agent **570**. The action rule can perform an action based on one or more parameter value of the action rule. In some embodiments, the action output of the action rule can be sent directly to the physical device, e.g., the building subsystems **122**. In some embodiments, the action output can be stored into the graph **529**. Another operating component of the building data platform **100**, e.g., the command processor **136**, can read the action from the graph **529** can communicate a corresponding command to the building subsystems **122**.

In some embodiments, instead of a digital twin triggering its own action, a trigger of the digital twin can trigger an action of another digital twin. For example, a rules engine of the twin manager **108** (e.g., the rules engine **926** described in FIGS. **9-10**) can link multiple triggers and actions of various digital twins together. For example, the rules engine **926** can store a logical rule with conditional requirements, e.g., triggers being tripped of one or more digital twins, Boolean logic and/or operations (e.g., AND, OR, XOR, NAND, NOT, XNOR, NOR, greater than, less than, equals, etc.), etc. The rule can indicate that when conditions of various triggers of various digital twins are satisfied, one or more actions of various digital twins should be executed.

Referring generally to FIGS. 12-23, systems and methods for using artificial intelligence to determine triggers and actions for an agent is shown. The triggers can trigger autonomously based on received data and cause an action to occur. In some embodiments, multiple digital twins can interact with each other by identifying interrelationships between each other via the graph 529, e.g., a VAV digital twin could interact with an AHU digital twin responsive to identifying that a VAV represented by the VAV digital twin is related to an AHU represented by the AHU digital twin via the graph 529. The digital twins can in some embodiments, simulate the impact of triggers and/or actions to validate and learn triggers and/or actions.

In some embodiments, the building data platform 100 can perform q-learning (Reinforcement Learning) to train and/or retrain the triggers and/or actions of the agents. In some embodiments, the data used to train and/or retrain the triggers and/or actions can be simulated data determined by another digital twin.

One digital twin may have trigger conditions such as, “when the outside temperature is x_0 ,” “when the inside humidity is $x\%$,” “when an AI-driven algorithm’s threshold is reached,” and “when it is a certain day of the week.” In responsive to one or multiple triggers being met, the digital twin can perform actions (e.g., capabilities of a device either inherent and/or digital twin enhanced). The actions can include setting a setpoint to a value x_0 . The actions can be to run a fan for x minutes. The actions can be to start an AI-driven energy saving schedule. The actions can be to change a mode status to an away status. In some embodiments, the building data platform 100 can use other digital twins to simulate a reward for various values of the triggers and/or actions. The reward can be optimized to determine values for the parameters of the triggers and/or actions.

In some embodiments, allowing the digital twin to learn and adjust the parameters of the triggers and/or rules allows the digital twin to optimize responses to internal and/or external events in real-time. In some embodiments, the digital twin performs operations with the correlation of contextual relationships to provide spatial intelligence. In some embodiments, the digital twin allows for AI-based self-learning solutions to operate on top of the digital twin. The digital twin can capture comprehensive data that drives rich analytics for security, compliance, etc. In some embodiments, the digital twin can enable and perform simulations.

In some embodiments, the building data platform 100 can identify events and/or event patterns if the building data platform 100 identifies a pattern that suggests a trigger and/or action should be updated. For example, if the building data platform 100 identifies a pattern occurring in a building, the building data platform 100 can set triggers and/or actions in digital twins to allow the pattern to occur automatically. For example, if a user closes their blinds at 5:00 P.M. regularly on weekdays, this could indicate that the user desires the blinds to be closed at 5:00 P.M. each day. The building data platform 100 can set a blind control digital twin to trigger a blind closing action at 5:00 P.M. each day.

In some embodiments, an agent of a digital twin can predict an inference in the future indicating that some action should be performed in the future. The building data platform 100 can identify that the action should be performed in the future and can set up a flow so that a prediction of one digital twin can be fed into another digital twin that can perform the action.

Referring now to FIG. 12, a system 1200 of a trigger rule 1202 of a thermostat digital twin where parameters of the trigger rule 1202 are trained is shown, according to an

exemplary embodiment. In some embodiments, the system 1200 can implement a model that rewards triggers and/or actions of the thermostat digital twin using a neural network that is trained from data aggregated from a related digital twin of the thermostat digital twin, an air handler unit digital twin.

The building data platform 100 can perturb parameters, ϵ_1 and ϵ_2 of the trigger rule 1202 of the thermostat digital twin. The trigger rule 1202 may be that if a number of occupants is greater than ϵ_1 and a zone temperature is less than ϵ_2° C. the rule is triggered and a corresponding action be performed. The corresponding action can be to increase a supply air temperature setpoint of an AHU to 22° C. The perturbation of the parameters can be increasing or decreasing the parameters in set amounts from existing values. The perturbation of the parameters can be selecting a space of values for the parameters and/or randomizing the parameters and/or parameter space.

With the perturbed values for ϵ_1 and ϵ_2 , the AHU digital twin 1204 can simulate the state of the AHU via the AHU digital twin 1204 for various conditions of occupant number and zone temperature. The result of the various states of the AHU digital twin 1204. The simulation can be performed by the AI agent 570 via the model 576. The output of the model 576 can be the simulated states, e.g., timeseries 566.

The building data platform 100 can analyze the states produced by the AHU digital twin 1204 to determine energy and comfort results from the states of the AHU digital twin 1204. For example, an energy score can be generated for each state. For example, a power consumption level can be determined for each state. Similarly, a comfort violation score can be determined for each state. The comfort violation can indicate whether or not a temperature, humidity, or other condition of a physical space controlled by the AHU would be uncomfortable for a user (e.g., go below or above certain levels).

The building data platform 100 can generate accumulated training data. The accumulated training data can include the values of the parameters ϵ_1 and ϵ_2 , the state of the AHU digital twin 1204 for each value of the parameters, and the energy score and comfort violation score for each state. In some embodiments, the triggers and/or actions that can be recommended for the thermostat digital twin can be determined by observing the responses of other digital twins on perturbed thresholds of existing triggers and/or actions.

The building data platform 100 can generate neural networks 1210 for predicting an energy score based on the parameters ϵ_1 and ϵ_2 . Furthermore, the neural networks 1210 can indicate a comfort violation score for the parameters ϵ_1 and ϵ_2 . The neural networks 1210 can be trained by the building data platform 100 based on the accumulated training data 1208.

Based on the trained neural network models 1210, the building data platform 100 can determine optimal values for the parameters ϵ_1 and ϵ_2 . The building data platform 100 can search a space of potential values for ϵ_1 and ϵ_2 that consider predicted energy scores and/or comfort violation scores predicted by the trained neural network models 1210. The optimization can be the relation 1400 shown in FIG. 14. The optimization 1212 performed by the building data platform 100 can be a method of computing the optimal threshold of a trigger conditions using the neural network models 1210 of rewards (e.g., energy and comfort) and solving constrained optimization model. Similarly, the optimization 1212 performed by the building data platform 100 to determine the

optimal threshold of action commands using the neural network models **1210** of rewards and solving constrained optimization.

In some embodiments, the optimal values for the parameters found by the building data platform **100** can be presented to a user for review and/or approval via a user interface, e.g., via the user device **176**. In some embodiments, the recommendations produced by the building data platform **100** through the components **1202-1212** can be restricted by only looking at state/value changes of digital twins that are nearest neighbors in the graph **529**, e.g., two nodes are directed related by one edge, e.g., a thermostat node for the thermostat digital twin is directed to an AHU node for the AHU digital twin **1204**. In some embodiments, the building data platform **100** can use spatial correlation to assume contextual relationship between assets that can affect each other's attribute states/values.

Referring now to FIG. **13**, a process **1300** for identifying values for the parameters of the trigger rule **1202** of FIG. **12** is shown, according to an exemplary embodiment. The process **1300** can be performed by the building data platform **100** and/or any component of the building data platform **100**. The process **1300** can be performed by the system **500** and/or components of the system **500**. Furthermore, the process **1300** can be performed by any computing device described herein.

In step **1302**, the building data platform **100** can perturb a thermostat digital twin (e.g., the thermostat digital twin rule **1202**) with various value for thresholds and/or other parameters, ϵ . The result of the perturbed parameters can result in various states, s . The states can be states predicted by the thermostat digital twin or another digital twin that operates based on the thresholds and/or parameters ϵ , e.g., the AHU digital twin **1204**. The perturbations and simulated states can result in pairs (S, ϵ) . The pairs can be used to determine feedback for energy and/or comfort, e.g., (E, C) .

In step **1304**, the building data platform **100** can building neural network models, e.g., the neural networks **1210** based on the data determined in step **1302**. The neural networks **1210** can predict energy rewards as a function of the state and the parameters, e.g., $E=f(s, \epsilon)$. Furthermore, the neural networks **1210** can predict comfort rewards as a function of the state and the parameters, e.g., $C=f(s, \epsilon)$.

In step **1306**, the building data platform **100** can determine a value for the parameter, ϵ that minimizes a relation, $(\alpha_1 \cdot E + \alpha_2 \cdot C)$. The minimization is shown in relation **1400** of FIG. **14**. The values of α_1 and α_2 can weigh the various rewards in the relation that is minimized, e.g., the energy reward and/or the comfort reward. In step **1308**, the building data platform **100** can periodically repeat the steps **1302-1306**. For example, the building data platform **100** can repeat the steps at a defined time period. In some embodiments, the building data platform **100** can compute rewards for the actions of the thermostat digital twin. If the rewards indicate that the thermostat digital twin need retraining, the building data platform **100** can repeat the steps **1302-1308**.

Referring now to FIG. **15**, a system **1500** of components where an action rule **1502** of a thermostat digital twin is shown where parameters of the action rule **1502** are trained, according to an exemplary embodiment. The system **1500** can include similar and/or the same components of FIG. **14**. The process **1300** of FIG. **13** can be applied to the action rule **1502** to train the parameters of the action rule **1502**.

The thermostat digital twin rule **1502** can be an action rule that if a trigger is met (e.g., the trigger **1402**), the action rule **1502** executes to command the AHU digital twin **1204**. The

trigger rule may be to execute the action rule if an occupant count is greater than ten and a zone temperature is less than twenty degrees Celsius. The action rule **1502** may be to increase an AHU supply air temperature setpoint to a value, e.g., ϵ . The value can, in some embodiments, be 22 degrees Celsius.

The building data platform **100** can predict states resulting from perturbed values of ϵ by executing the AHU digital twin **1204** to simulate the states. The building data platform **100** can collect rule feedback **1206** to construct accumulated training data **1208**. Furthermore, the building data platform **100** can train neural network models **1210** based on the accumulated training data **1208** and find optimal values for the parameter ϵ based on the trained neural network models **1210**.

Referring now to FIG. **16**, a list **1600** and a list **1602** of states of a zone and of an air handler unit that can be used to train the parameters of the trigger rule and the action rule of the thermostat digital twins of FIGS. **12-15** is shown, according to an exemplary embodiment. The list **1600** includes states for a zone. The states can be zone temperature, zone humidity, outdoor air temperature, outdoor air humidity, zone occupancy, etc. These states can be predicted and/or determined based on a digital twin for a space based on perturbed parameter values for a trigger rule, an action rule, weather forecasts, etc. In this regard, the rule feedback **1206**, in some embodiments, can be generated based on the digital twin for the space and used to tune the values of the parameters for the trigger rule **1402** and/or the action rule **1502**.

The list **1602** includes states for an AHU. The states can be supply air temperature, supply air flow rate, return air temperature, return air flow rate, outdoor air flow rate, etc. These states can be predicted and/or determined based on a digital twin for an AHU (e.g., the AHU digital twin **1204**) based on perturbed parameter values for a trigger rule, an action rule, etc. In this regard, the rule feedback **1206** in some embodiments, can be generated based on the digital twin for the AHU and used to tune the values of the parameters for the trigger rule **1402** and/or the action rule **1502**.

Referring now to FIG. **17**, a system **1700** of a trigger rule of a chemical reactor digital twin where parameters of a trigger rule are trained is shown, according to an exemplary embodiment. A reactor feed digital twin which may model the feed of a chemical reactor can include various trigger rules and/or action rules, e.g., the trigger rule **1702**. The trigger rule **1702** can be that if a chemical concentration of a first chemical A is less than ϵ_1 (e.g., 10 g/l) and a chemical concentration of a second chemical B is less than ϵ_2 (e.g., 20 g/l) then an action rule is triggered. The action rule may be increase a catalyst C feed amount to 300 g/s.

The building data platform **100** can perturb the values for the parameters ϵ_1 and ϵ_2 of the reactor feed digital twin trigger rule **1702** (e.g., pseudo-randomly, increasing and/or decreasing in a particular number of predefined increments, etc.). A chemical reactor digital twin **1704** can simulate a state of the chemical reactor for the various perturbed parameters ϵ_1 and ϵ_2 . The building data platform **100** can determine a rule feedback **1706** for the state simulate by the chemical reactor digital twin **1704**. The rule feedback **1706** can identify scores for production throughput (P) and chemical property (C).

The building data platform **100** can accumulate training data **1708**. The accumulated training data **1708** can include the feedback **1706**, the state simulated by the chemical reactor digital twin **1704**, and/or the parameter values for ϵ_1

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and ε_2 . The building data platform 100 can train neural network models 1710 to predict production throughput and/or chemical property for the various parameter and/or state pairs, e.g., the state resulting from the parameters of the trigger rule 1702. The building data platform 100 can use the trained neural network models 1710 to identify optimal values for ε_1 and ε_2 . In element 1712, the building data platform 100 can identify values for ε_1 and ε_2 that minimize the relation 1900 shown in FIG. 19. In some embodiment, the optimization can optimize production throughput and/or chemical property.

Referring now to FIG. 18, a process 1800 for identifying values for the parameters of the trigger rule of FIG. 17 is shown, according to an exemplary embodiment. The process 1800 can be performed by the building data platform 100 and/or any component of the building data platform 100. The process 1900 can be performed by the system 500 and/or components of the system 500. Furthermore, the process 1900 can be performed by any computing device described herein. The steps 1802-1808 can be the same as or similar to the steps 1302-1308. However, the steps 1802-1808 can be executed for a reactor digital twin and the reward for training the neural networks and be production throughput and chemical property.

In step 1802, the building data platform 100 can perturb a reactor digital twin 1704 with various values of a threshold ε of a trigger rule 1702 with various values which cause the reactor digital twin to determine resulting states for the various values of the threshold, ε . The states and the values for the threshold ε can create state threshold pairs. The pairs can be used to determine feedback, e.g., production throughput and chemical property.

In step 1804, after some accumulation of feedback data, the building data platform 100 can build neural network models 1710 based on the pairs that predict production throughput and chemical property based on the values for the threshold ε . In step 1806, the building data platform 100 can determine a value for the threshold ε that maximizes a reward and/or minimizes a penalty. The building data platform 100 can minimize the relation 1900 of FIG. 19. In step 1808, the building data platform 100 can periodically retrain the values for the threshold ε for the trigger rule 1702.

Referring now to FIG. 20, a system 2000 including an action rule 2002 of a chemical reactor digital twin where parameters of the action rule 2002 are trained is shown, according to an exemplary embodiment. The reactor feed twin rule 2002 can be an action rule to increase a catalyst C feed amount to ε_1 g/s in response to an trigger rule being triggered, e.g., the trigger rule 1702. The building data platform 100 can perturb the values of the parameter ε_1 and the reactor digital twin 1704 can predict states resulting from the perturbed parameter. The building data platform 100 can determine rule feedback 1706 and generate accumulated training data 1708 based on the rule feedback 1706. The building data platform 100 can train the neural network models 1710. Based on the neural network models 1710, the building data platform 100 can find optimal values for the parameter ε_1 .

Referring now to FIG. 21, a list 2100 and a list 2102 of states of a feed of a reactor and a reactor that can be included in the trigger rule and the action rule of FIGS. 12-15 are shown, according to an exemplary embodiment. The list 2100 includes states for a feed of a chemical reactor. The states can be reactants feed amount, catalysts feed amount, feed stream temperature, etc. These states can be predicted and/or determined based on a digital twin for a space based on perturbed parameter values for a trigger rule, an action

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rule, etc. In this regard, the rule feedback 1706 in some embodiments, can be generated based on the digital twin for the space and used to tune the values of the parameters for the trigger rule 1702 and/or the action rule 2002.

The list 2102 includes states for a chemical reactor. The states can be product concentration, cooling coil temperature, product temperature, etc. These states can be predicted and/or determined based on a digital twin for a chemical reactor (e.g., the reactor digital twin 1704) based on perturbed parameter values for a trigger rule, an action rule, etc. In this regard, the rule feedback 1706 in some embodiments, can be generated based on the digital twin for the chemical reactor and used to tune the values of the parameters for the trigger rule 1702 and/or the action rule 2002.

Referring now to FIG. 22, a system 2200 where triggers and actions that can be constructed and learned for a digital twin is shown, according to an exemplary embodiment. Considering a building where a room in the building has a thermostat, the building data platform 100 can construct triggers and/or actions of an agent of a digital twin or the room. The triggers and/or actions can be determined with an energy reduction reward function 2204 by a learning service 2206. The energy reduction reward function 2204 can produce triggers and/or actions that have values that minimize energy usage.

In some embodiments, the building data platform 100 can search the graph 529 to identify information related to the space, e.g., related pieces of equipment, spaces, people, etc. For example, the building data platform 100 can identify which entities of the graph 529 are related and operate to affect each other. The building data platform 100 can identify which actions each entity can perform and/or what measurements each entity can make, e.g., by identifying related data nodes for each entity. The identified entities, measurements, and/or commands can be combined into the rule 2202 by the building data platform 100.

In some embodiments, the learning service 2206, which may be a component of the building data platform 100, can run a learning process with the rule 2202 and/or one or more reward functions (e.g., comfort reward function, carbon footprint reduction reward function, the energy reduction reward function 2204, etc.). The learning service 2206 can learn the rule 2208 from the rule 2202 and/or the energy reduction reward function 2204.

The learning service 2206 can run an optimization to determine combinations between measurements and actions triggered based on the measurements. The learning service 2206 can determine values for each measurement and/or action. Furthermore, the learning service 2206 can identify the relational operations for causing a trigger, e.g., equals to, greater than, less than, not equal to, etc. Furthermore, the learning service 2206 can identify action operations, e.g., increase by a particular amount, decrease by a particular amount, set an output equal to a value, run a particular algorithm, etc.

Referring now to FIG. 23, a process 2300 for constructing triggers and actions for a digital twin is shown, according to an exemplary embodiment. In some embodiments, the process 2300 can be performed by the building data platform 100. In some embodiments, the process 2300 can be performed by the learning service 2206.

In step 2302, the building data platform 100 can determine actions that a particular entity can take and data that the entity can measure by analyzing a graph 529. The entity can be a thermostat, an air handler unit, a zone of a building, a person, a VAV unit, and/or any other entity. For example, if the entity is a thermostat the building data platform 100

could identify room temperature measurements for a thermostat and/or a cooling stage command, a heating stage command, a fan command, etc. that the thermostat can perform. Responsive to identifying data that the entity can measure, the building data platform **100** can generate a trigger condition based on the data type, e.g., when the temperature is equal to, less than, greater than, and/or not equal to some parameter value, trigger an action.

In step **2304**, the building data platform **100** identifies, based on the graph **529**, entities related to the entity and actions that the entities can take and data that the entities can measure. For example, if the entity is for a thermostat for a zone, the building data platform **100**, could identify a shade control system for controlling a shade of the zone, an air handler unit that serves the zone, a VAV that serves the zone, etc. For example, the building data platform **100** can identify, based on the building graph **529**, that a binds node is associated with a zone node that the thermostat node is related to. The building data platform **100** can identify a list of actions that the entities can perform, e.g., setting blind position from 0% (fully open) to 100% (fully closed).

In some **2306**, the building data platform **100** can simulate various combinations of triggers that are based on the data that the entity and/or entities can measure and actions that are based on the actions that the entity and/or entities can make. The building data platform **100** can simulate various combinations, trigger operations, action operations, and/or parameters.

In step **2308**, the building data platform **100** can identify a combination of triggers and actions that maximizes a reward. The building data platform **100** can search the simulated combinations of triggers and/or actions to identify a trigger and/or action that maximizes a reward and/or minimizes a reward. In some embodiments, the building data platform **100** uses a policy gradient and value function instead of brute force to try out combinations of the triggers and/or actions in the steps **2306-2308**.

In some embodiments, the building data platform **100** can identify the operations for the triggers and/or actions. For example, the operation could be comparing a measurement to a threshold, determining whether a measurement is less than a threshold, determining whether a measurement is greater than the threshold, determining whether the measurement is not equal to the threshold, etc.

In step **2310**, the building data platform **100** can generate a digital twin for the entity. The entity can include (or reference) the graph **529** and include an agent that operates the triggers and/or actions. The triggers and/or actions can operate based on the graph **529** and/or based on data received building equipment, e.g., the building subsystems **122**.

In step **2312**, the building data platform **100** can run a building system of a building and monitor the behavior of the entity and entities of the building. In some embodiments, the building system can be the building subsystems **122**. In step **2314**, the building data platform **100** can identify relationships between the measurements and actions of the entity and/or the entities based on the monitored behavior. The building data platform **100** can discover existing relationships by identifying how the measurements are currently affecting actions based on the monitored behavior. In step **2316**, the building data platform **100** can optimize the identified relationships between the measurements and the actions by maximizing a reward or minimizing a penalty.

Configuration of Exemplary Embodiments

The construction and arrangement of the systems and methods as shown in the various exemplary embodiments

are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

In various implementations, the steps and operations described herein may be performed on one processor or in a combination of two or more processors. For example, in some implementations, the various operations could be performed in a central server or set of central servers configured to receive data from one or more devices (e.g., edge computing devices/controllers) and perform the opera-

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tions. In some implementations, the operations may be performed by one or more local controllers or computing devices (e.g., edge devices), such as controllers dedicated to and/or located within a particular building or portion of a building. In some implementations, the operations may be performed by a combination of one or more central or offsite computing devices/servers and one or more local controllers/computing devices. All such implementations are contemplated within the scope of the present disclosure. Further, unless otherwise indicated, when the present disclosure refers to one or more computer-readable storage media and/or one or more controllers, such computer-readable storage media and/or one or more controllers may be implemented as one or more central servers, one or more local controllers or computing devices (e.g., edge devices), any combination thereof, or any other combination of storage media and/or controllers regardless of the location of such devices.

What is claimed:

1. A building system comprising one or more storage devices storing instructions thereon that, when executed by one or more processors, cause the one or more processors to:
 - store a data structure of a digital twin of an entity of a building, the data structure storing information associated with at least one of the entity or a plurality of entities of the building;
 - execute a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm;
 - store the trigger rule and the action rule in the digital twin;
 - determine that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule;
 - determine an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule; and
 - cause one or more devices to operate based on the action determined by the action rule.
2. The building system of claim 1, wherein the instructions cause the one or more processors to:
 - provide the action to one or more other digital twins;
 - determine that the action triggers one or more trigger rules of the one or more other digital twins; and
 - determine one or more actions of one or more action rules of the one or more other digital twins by executing the one or more action rules responsive to the action triggering the one or more trigger rules.
3. The building system of claim 1, wherein the data structure is a graph data structure comprising a plurality of nodes and a plurality of edges between the plurality of nodes, wherein the plurality of nodes represent the entity, the plurality of entities, and data points of the entity and the plurality of entities;
 - wherein the plurality of edges represent relationships between the entity, the plurality of entities, and the data points of the entity and the plurality of entities.
4. The building system of claim 1, wherein the instructions cause the one or more processors to cause the one or more devices to operate based on the action by:
 - sending a command to the one or more devices to perform the action; or

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store the action in the data structure and retrieve the action from the data structure and send the command based on the action retrieved from the data structure to the one or more devices.

5. The building system of claim 1, wherein the instructions cause the one or more processors to learn one or more parameters of the trigger rule or the action rule by:
 - adjusting a value of the one or more parameters of the trigger rule or the action rule to a plurality of different values;
 - simulating one or more states resulting from the plurality of different values of the one or more parameters of the trigger rule or the action rule; and
 - determine one or more updated values for the one or more parameters of the trigger rule or the action rule based on the one or more states resulting from the plurality of different values of the one or more parameters of the trigger rule or the action rule.
6. The building system of claim 5, wherein the instructions cause the one or more processors to determine the one or more updated values for the one or more parameters of the trigger rule or the action rule by:
 - determining a particular reward for the one or more states resulting from the plurality of different values of the one or more parameters of the trigger rule or the action rule;
 - training a model that predicts a reward based on the particular reward for the one or more states and the plurality of different values of the one or more parameters of the trigger rule or the action rule; and
 - performing an optimization that determines the one or more updated values for the one or more parameters of the trigger rule or the action rule based on the model and the reward.
7. The building system of claim 6, wherein the instructions cause the one or more processors to generate construct the trigger rule and the action rule based on actions and the information associated with the entity and actions and the information associated with the plurality of entities by:
 - determining trigger rules for the information associated with the entity and the information associated with the plurality of entities;
 - determining action rules for the actions of the entity and the actions of the plurality of entities;
 - simulating combinations of the trigger rules and the action rules, and determining rewards for the combinations; and
 - selecting the trigger rule and the action rule from the combinations of the trigger rules and the action rules based on the rewards of the combinations.
8. The building system of claim 6, wherein the instructions cause the one or more processors to construct the trigger rule and the action rule based on actions and the information associated with the entity and actions and the information associated with the plurality of entities by:
 - determining trigger rules for the information associated with the entity and the information associated with the plurality of entities, the trigger rules including the trigger rule;
 - determining action rules for the actions of the entity and the actions of the plurality of entities, the action rules including the action rule;
 - monitoring behavior of the building system of the building;
 - determining a relationship between the trigger rule of the trigger rules and the action rule of the action rules indicated by the behavior of the building system; and

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selecting the trigger rule from the trigger rules and the action rule from the action rules based on the relationship.

9. A method comprising:

storing, by a processing circuit, a data structure of a digital twin of an entity, the data structure storing information associated with at least one of the entity or a plurality of entities;

executing, by the processing circuit, a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm;

storing, by the processing circuit, the trigger rule and the action rule in the digital twin;

determining, by the processing circuit, that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule;

determining, by the processing circuit, an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule; and

causing, by the processing circuit, one or more devices to operate based on the action determined by the action rule.

10. The method of claim 9, further comprising:

providing, by the processing circuit, the action to one or more other digital twins;

determining, by the processing circuit, that the action triggers one or more trigger rules of the one or more other digital twins; and

determining, by the processing circuit, one or more actions of one or more action rules of the one or more other digital twins by executing the one or more action rules responsive to the action triggering the one or more trigger rules.

11. The method of claim 9, wherein the data structure is a graph data structure comprising a plurality of nodes and a plurality of edges between the plurality of nodes, wherein the plurality of nodes represent the entity, the plurality of entities, and data points of the entity and the plurality of entities;

wherein the plurality of edges represent relationships between the entity, the plurality of entities, and the data points of the entity and the plurality of entities.

12. The method of claim 9, wherein causing, by the processing circuit, the one or more devices to operate based on the action comprises:

sending a command to the one or more devices to perform the action; or

storing the action in the data structure and retrieve the action from the data structure and send the command based on the action retrieved from the data structure to the one or more devices.

13. The method of claim 9, further comprising learning, by the processing circuit, one or more parameters of the trigger rule or the action rule by:

adjusting a value of the one or more parameters of the trigger rule or the action rule to a plurality of different values;

simulating one or more states resulting from the plurality of different values of the one or more parameters of the trigger rule or the action rule; and

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determining one or more updated values for the one or more parameters of the trigger rule or the action rule based on the one or more states resulting from the plurality of different values of the one or more parameters of the trigger rule or the action rule.

14. The method of claim 9, further comprising generating, by the processing circuit, the digital twin by:

identifying actions and data associated with the entity based on the data structure;

identifying that the plurality of entities are related to the entity based on the data structure;

identifying actions and data associated with the plurality of entities based on the data structure; and

generating the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the plurality of entities.

15. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising:

storing a data structure of a digital twin of an entity of a building, the data structure storing information associated with at least one of the entity or a plurality of entities of the building;

executing a learning algorithm to construct a trigger rule and an action rule with the data structure of the digital twin as an input to the learning algorithm;

storing the trigger rule and the action rule in the digital twin;

determining that the trigger rule of the digital twin is triggered by comparing at least some of the information of the data structure against one or more conditions of the trigger rule by querying the data structure for data values and comparing the data values against the one or more conditions of the trigger rule;

determining an action of the action rule of the digital twin responsive to determining that the trigger rule is triggered by executing the action rule; and

causing one or more devices to operate based on the action determined by the action rule.

16. The one or more non-transitory computer-readable media of claim 15, the operations further comprising:

identifying that a relationship exists between the trigger rule and the action rule; and

determining the action of the action rule responsive to determining that the trigger rule is triggered and that the relationship exists between the trigger rule and the action rule.

17. The one or more non-transitory computer-readable media of claim 15, the operations further comprising:

storing a second digital twin associated with a second entity of the building, the second digital twin comprising a second action rule;

identifying a relationship between the trigger rule of the digital twin and the action rule of the second digital twin; and

determining a second action based on the second action rule of the second digital twin responsive to determining that the trigger rule of the digital twin is triggered.

18. The one or more non-transitory computer-readable media of claim 15, the operations further comprising:

receiving first timeseries data from the entity;

receiving second timeseries data from a second entity of the building separate from the entity; and

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determining that the trigger rule of the digital twin is triggered by comparing the first timeseries data and the second timeseries data against the one or more conditions of the trigger rule.

19. The one or more non-transitory computer-readable media of claim 15, the operations to generate the digital twin further comprising:

- identifying actions and data associated with the entity based on the data structure;
- identifying that the plurality of entities are related to the entity based on the data structure;
- identifying actions and data associated with the plurality of entities based on the data structure; and
- generating the trigger rule and the action rule based on the actions and the data associated with the entity and the actions and the data associated with the plurality of entities.

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20. The one or more non-transitory computer-readable media of claim 15, the operations further cause the one or more processors to construct the trigger rule and the action rule based on actions and the information associated with the entity and actions and the information associated with the plurality of entities by:

- determining trigger rules for the information associated with the entity and the information associated with the plurality of entities;
- determining action rules for the actions of the entity and the actions of the plurality of entities;
- simulating combinations of the trigger rules and the action rules, and determining rewards for the combinations; and
- selecting the trigger rule and the action rule from the combinations of the trigger rules and the action rules based on the rewards of the combinations.

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